[ADS5034-01] Computer Vision

09. Stereo Vision

Jae-Young Choi

Sungkyunkwan University

Outline

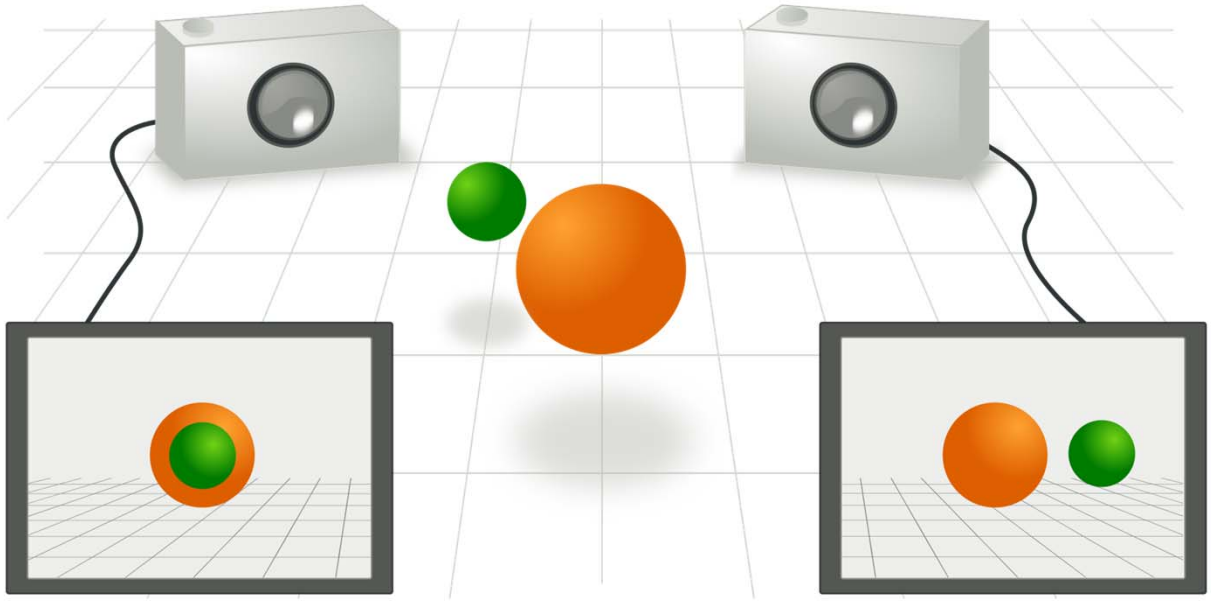


- Two-view Geometry
  - Epipolar Geometry
  - Essential matrix
  - Fundamental matrix
- Stereo Vision
  - Stereo Rectification
  - Stereo Matching
  - Disparity & Depth

Two-view Geometry

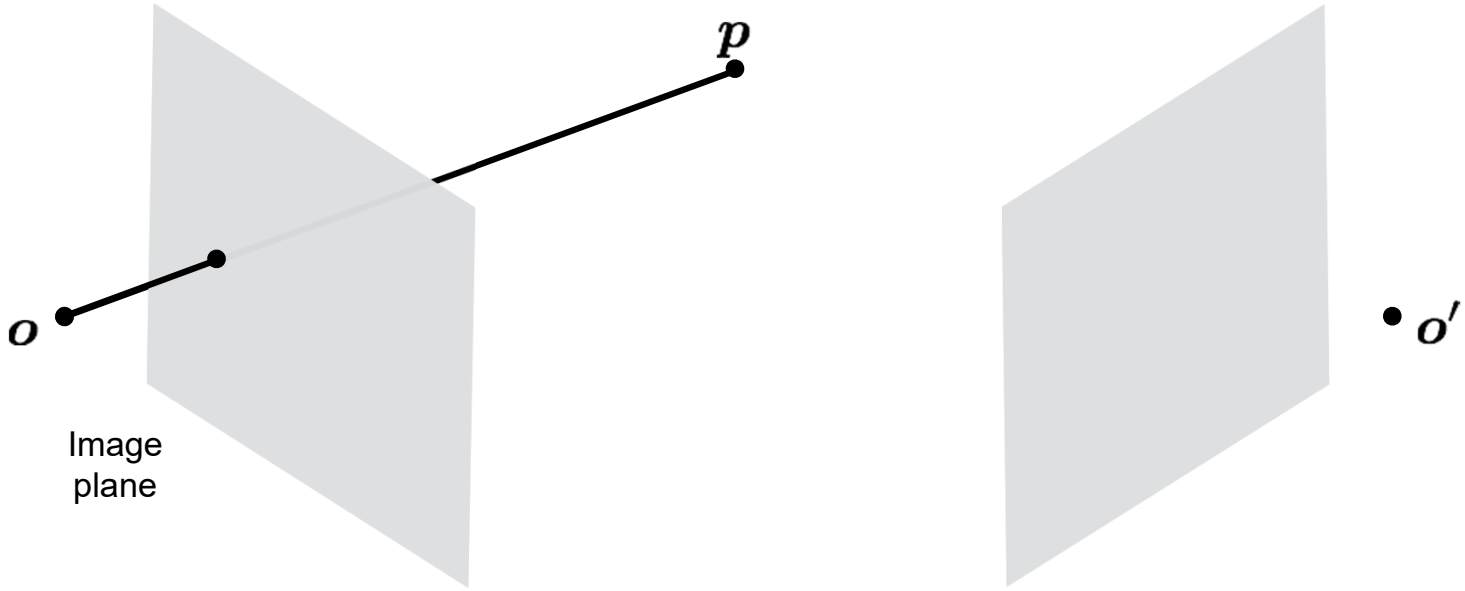
Epipolar Geometry

- Typical use case for epipolar geometry
  - Two cameras take a picture of the same scene from different points of view
  - The epipolar geometry then describes the relation between the two resulting views

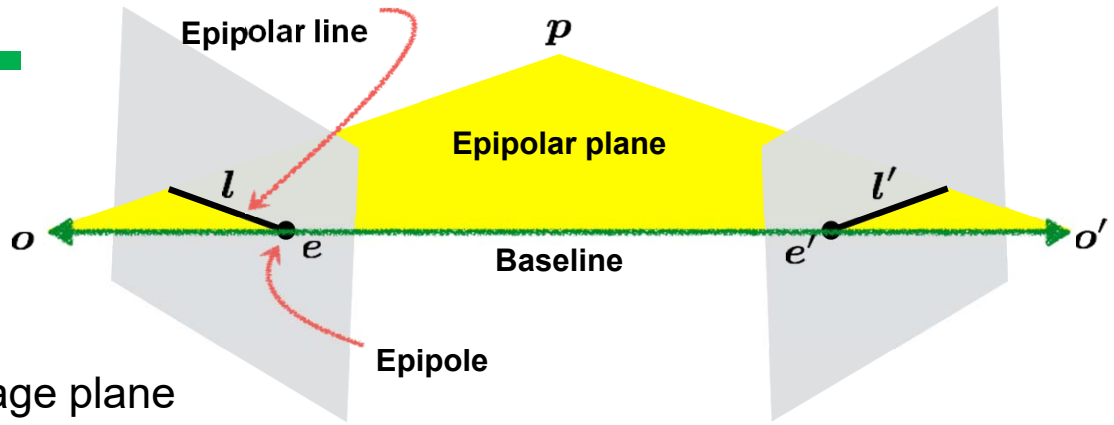


https://en.wikipedia.org/wiki/Epipolar_geometry

Epipolar Geometry



Epipolar Geometry

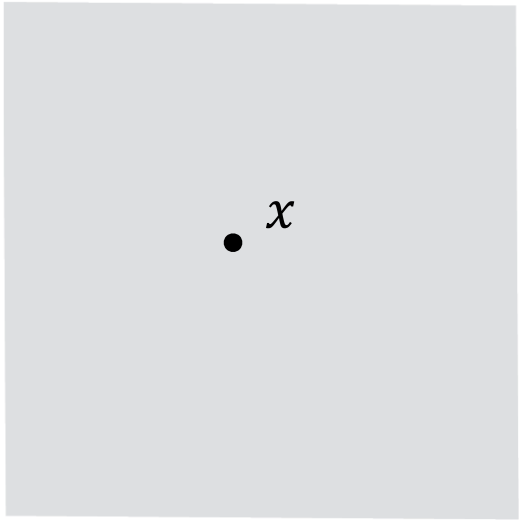


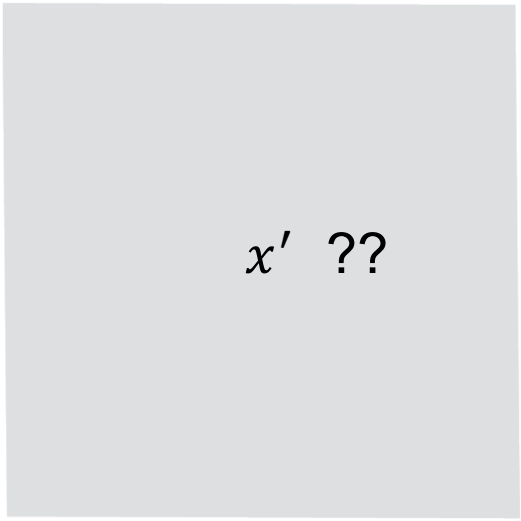
- Baseline
  - Line joining the camera centers
- Epipole
  - Point of intersection of baseline with the image plane
  - projection of o' on the image plane
- Epipolar plane
  - Plane containing baseline and world point
- Epipolar line
  - Intersection of epipolar plane with the image plane

- All epipolar lines intersect at the epipole
- An epipolar plane intersects the left and right image planes in epipolar lines

Epipolar Geometry

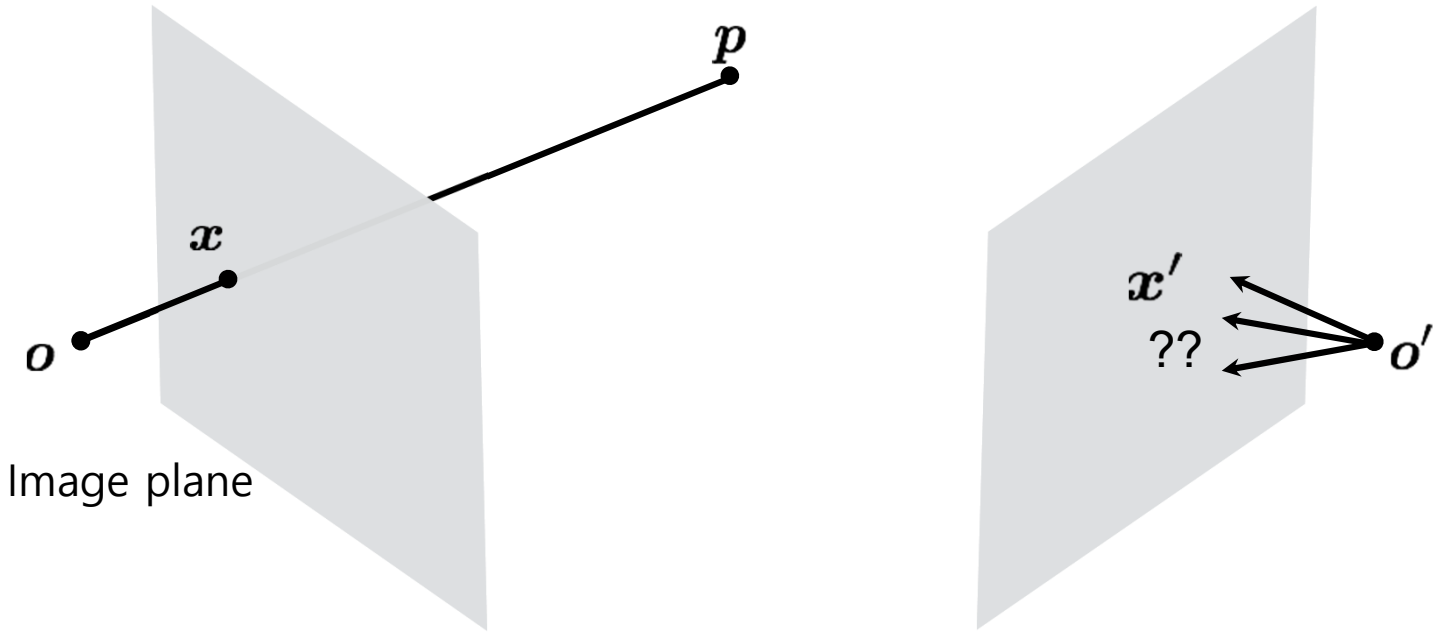
- Given x in left image, where can corresponding point x' be?





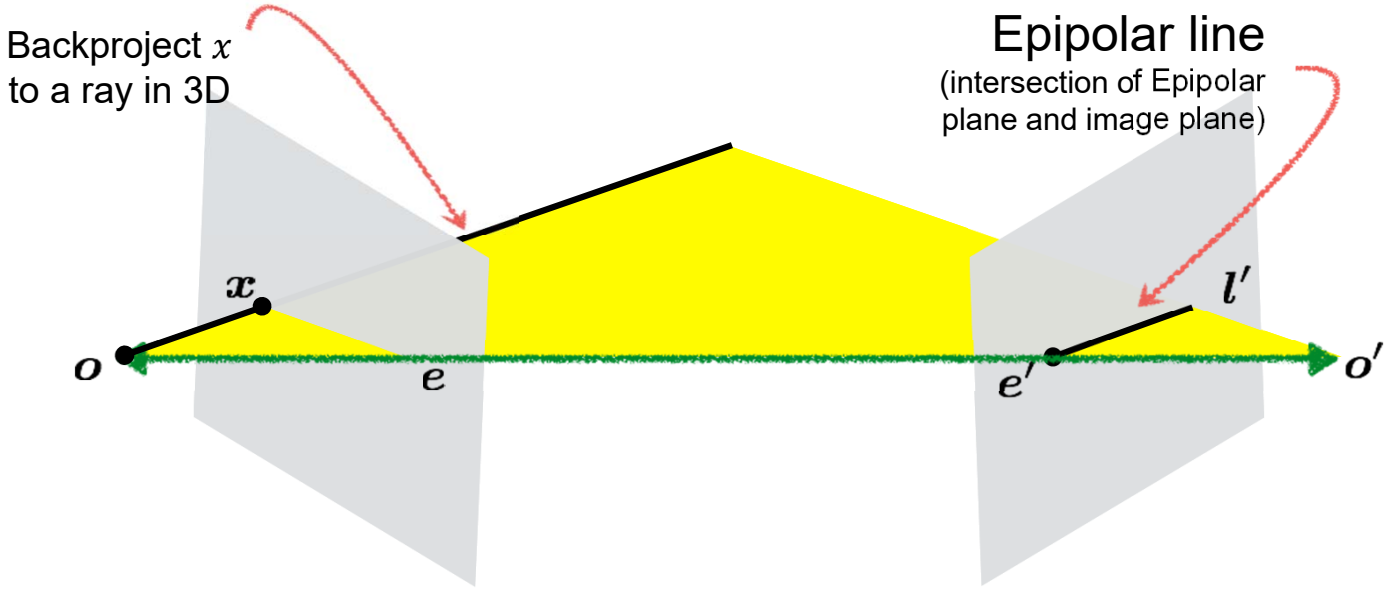
Epipolar Geometry

- Given x in left image, where can corresponding point x' be?



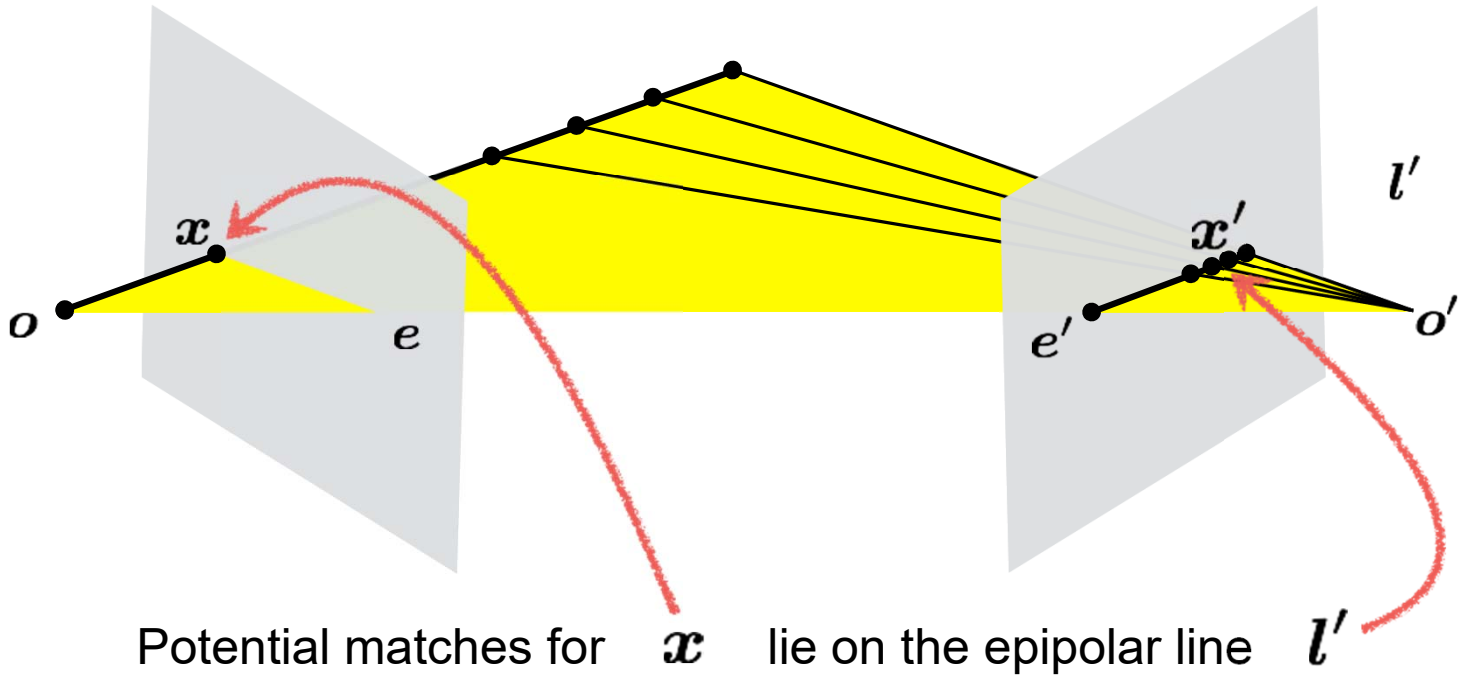
Epipolar Geometry

- Epipolar constraint
  - Geometry of two views allows us to constrain where the corresponding pixel for some image point in the first view must occur in the second view
  - Reduces correspondence problem to 1D searching along conjugate epipolar lines



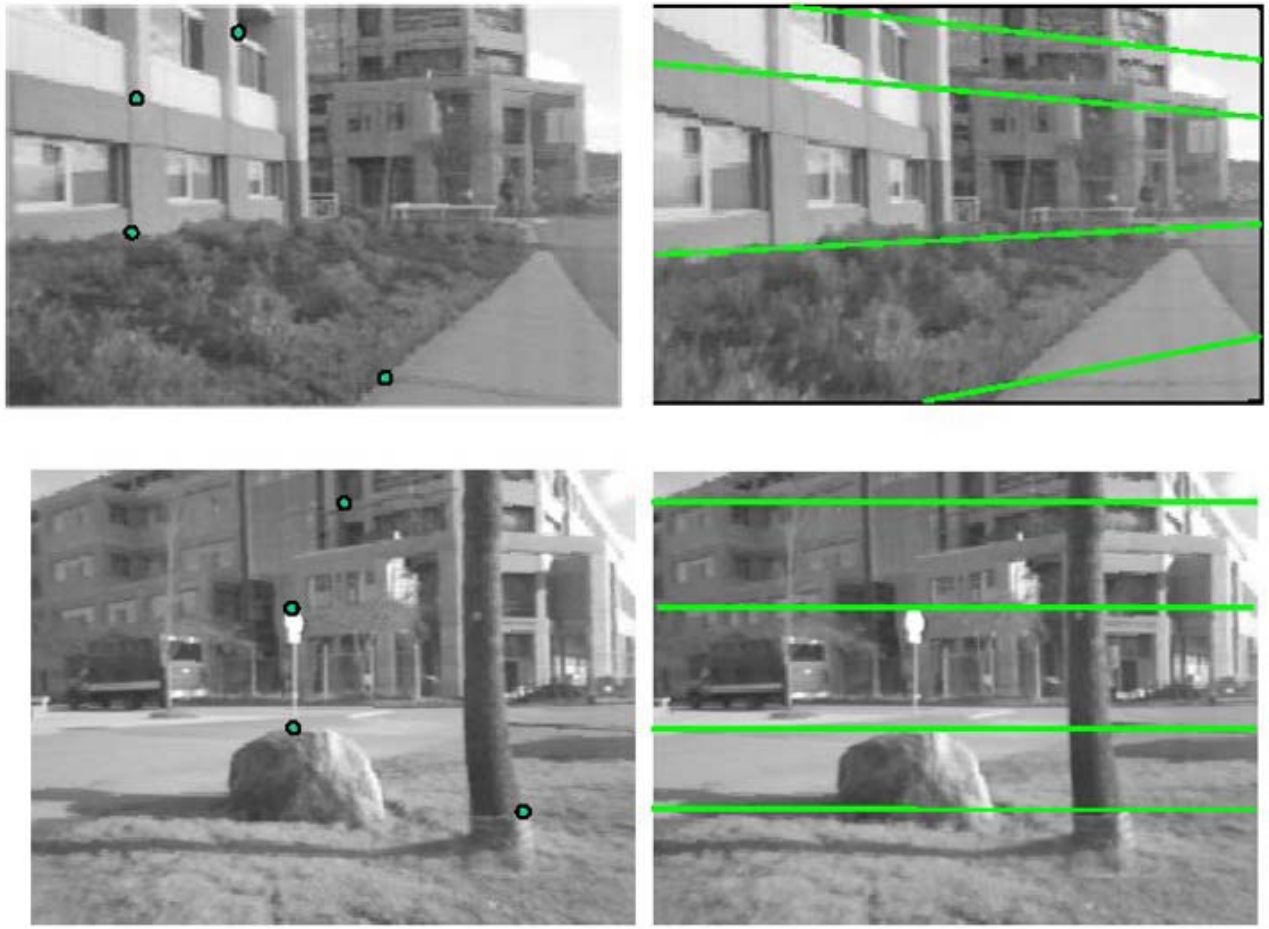
Epipolar Geometry

- Epipolar constraint



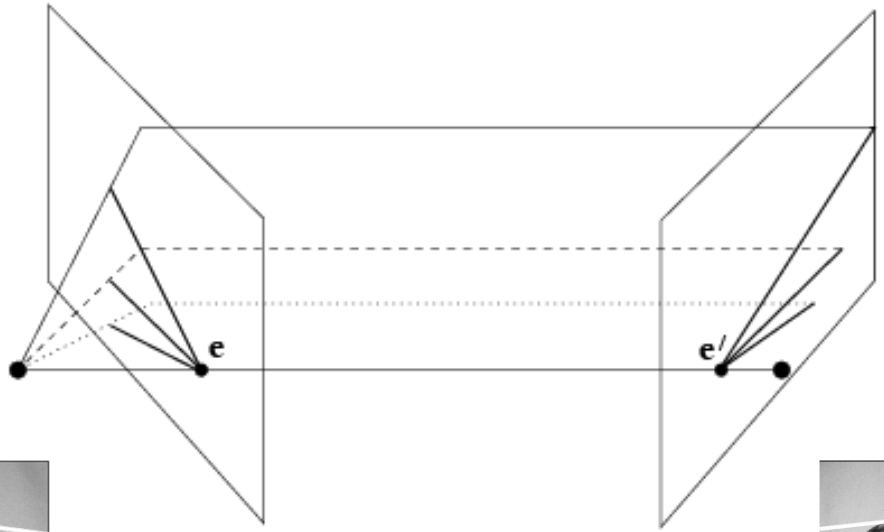
Epipolar Geometry

- Example

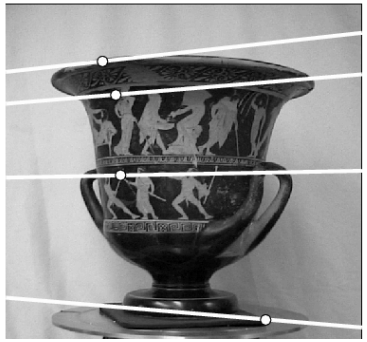


Epipolar Geometry

- Ex) Converging cameras



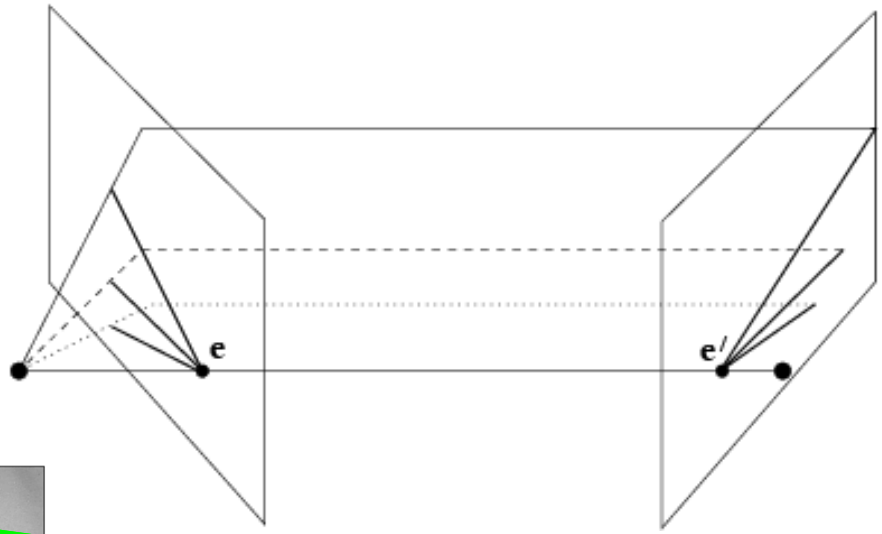




Where is the epipole in this image?

Epipolar Geometry

- Converging cameras





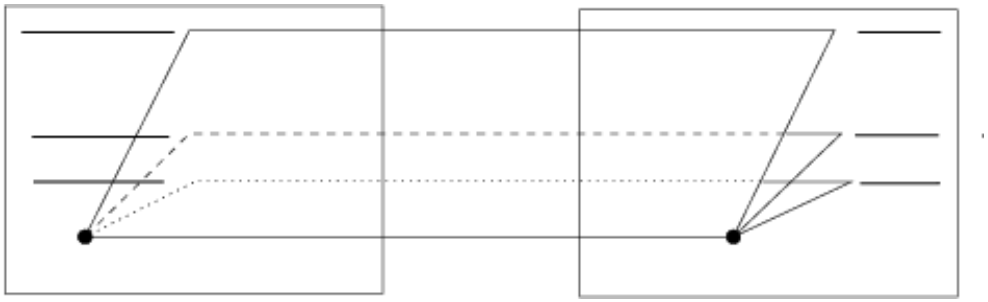
here!

It's not always in the image

Where is the epipole in this image?

Epipolar Geometry

- Parallel cameras



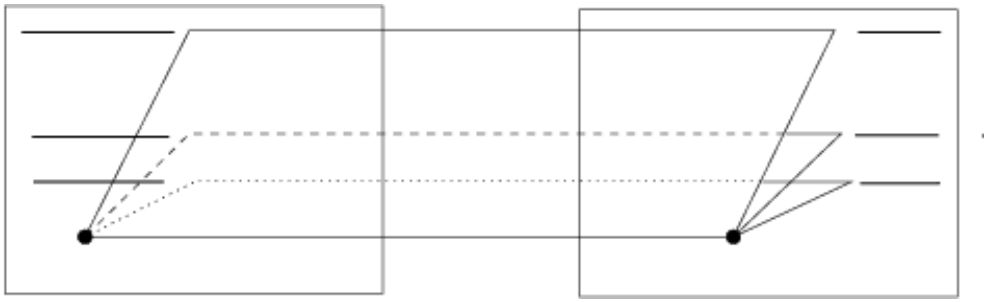


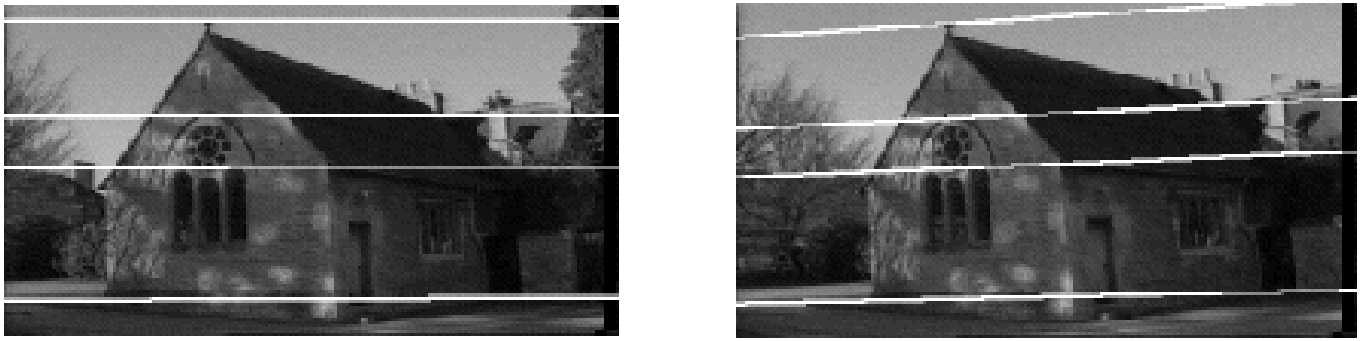


Where is the epipole?

Epipolar Geometry

- Parallel cameras



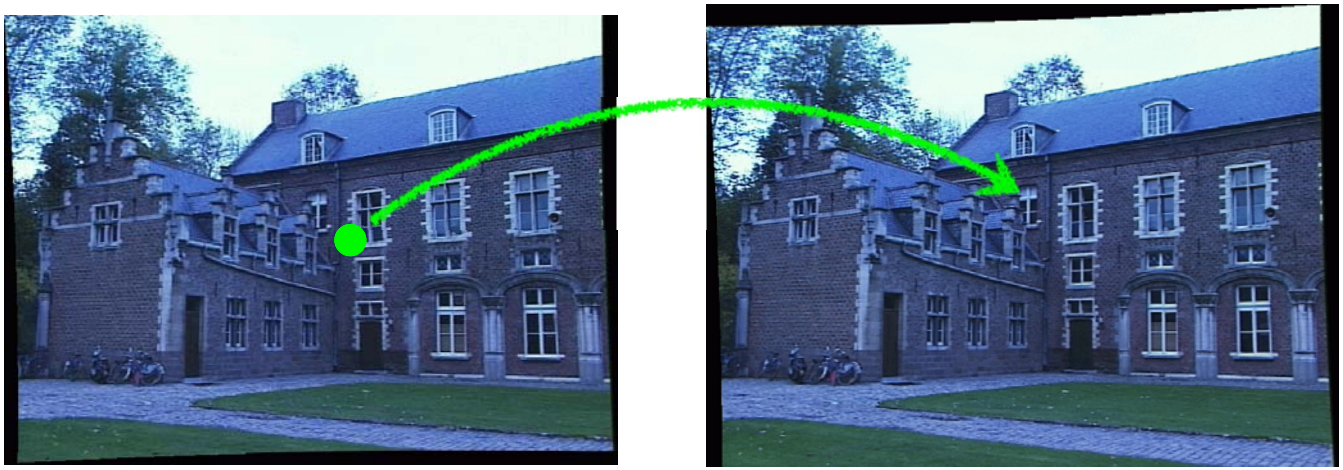


epipole at infinity

Epipolar Geometry

- The epipolar constraint is an important concept for stereo vision

Task: Match point in left image to point in right image

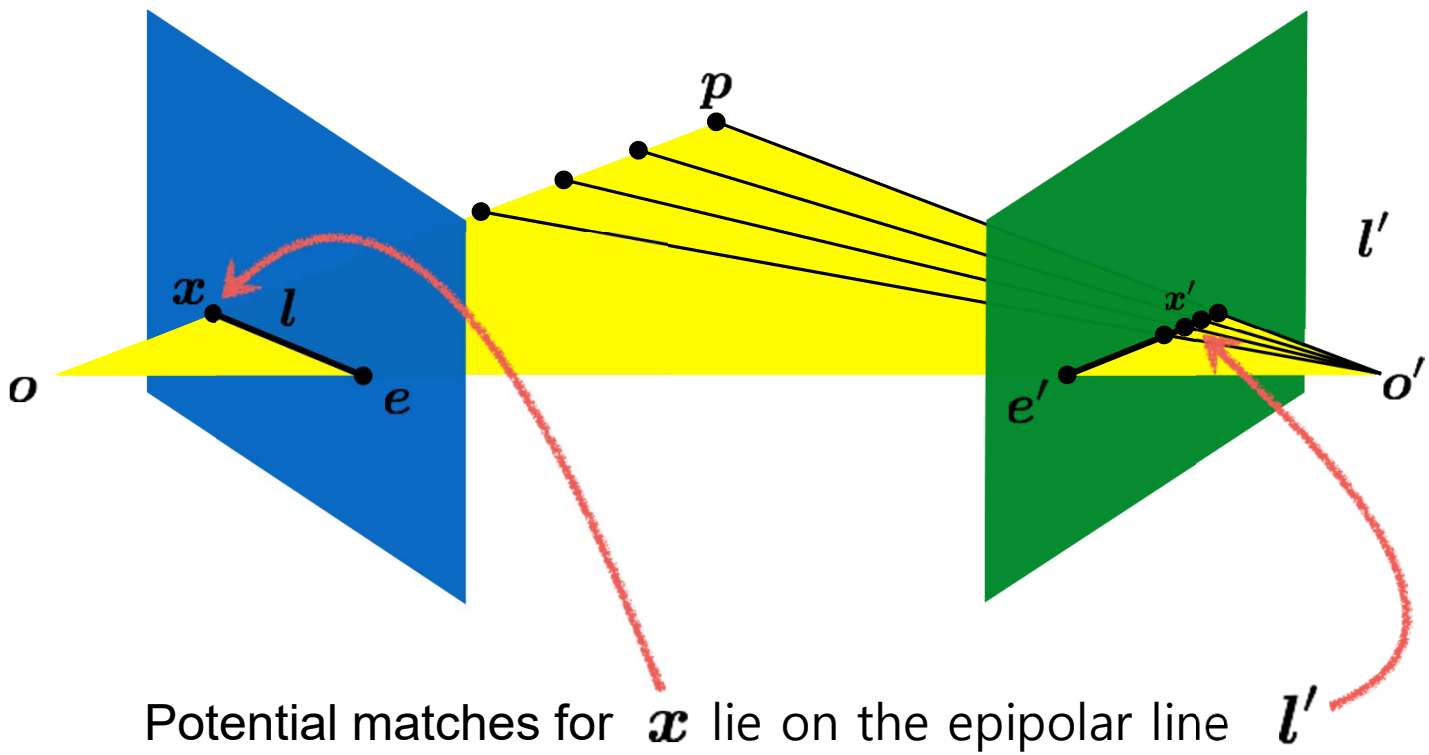


Left image

Right image

Epipolar Geometry

- Recall : Epipolar constraint

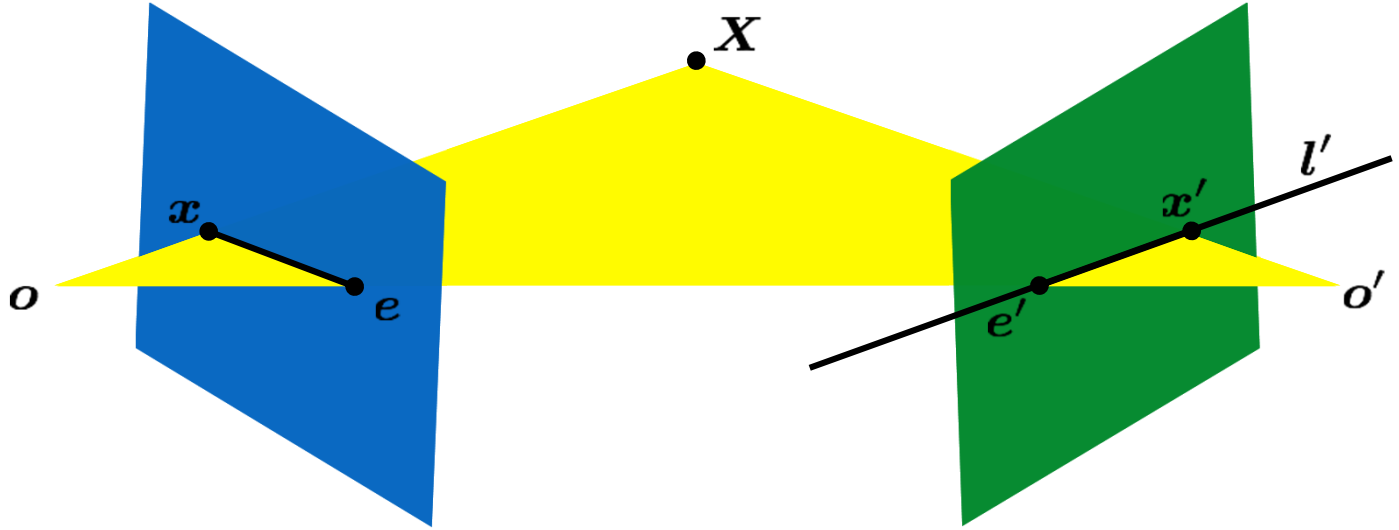


Potential matches for x lie on the epipolar line l'

Essential Matrix

- Given a point in one image, multiplying by the **essential matrix** will tell us the **epipolar line** in the second image

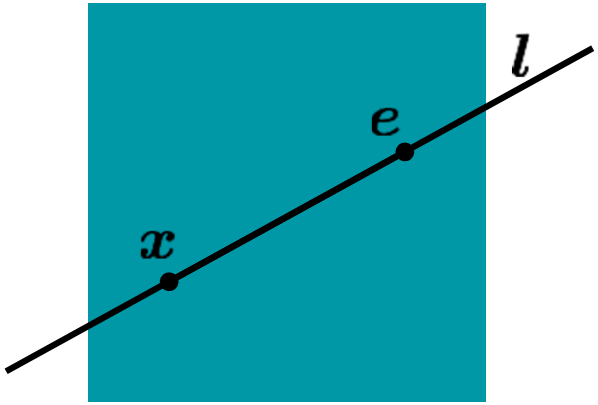
$$\mathbf{E}x = l'$$



Essential Matrix

- Epipolar Line

$$ax + by + c = 0$$ in vector form $$\boldsymbol{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$



If the point $\boldsymbol{x}$ is on the epipolar line $\boldsymbol{l}$ then

$$\boldsymbol{x}^\top \boldsymbol{l} = \ ?$$

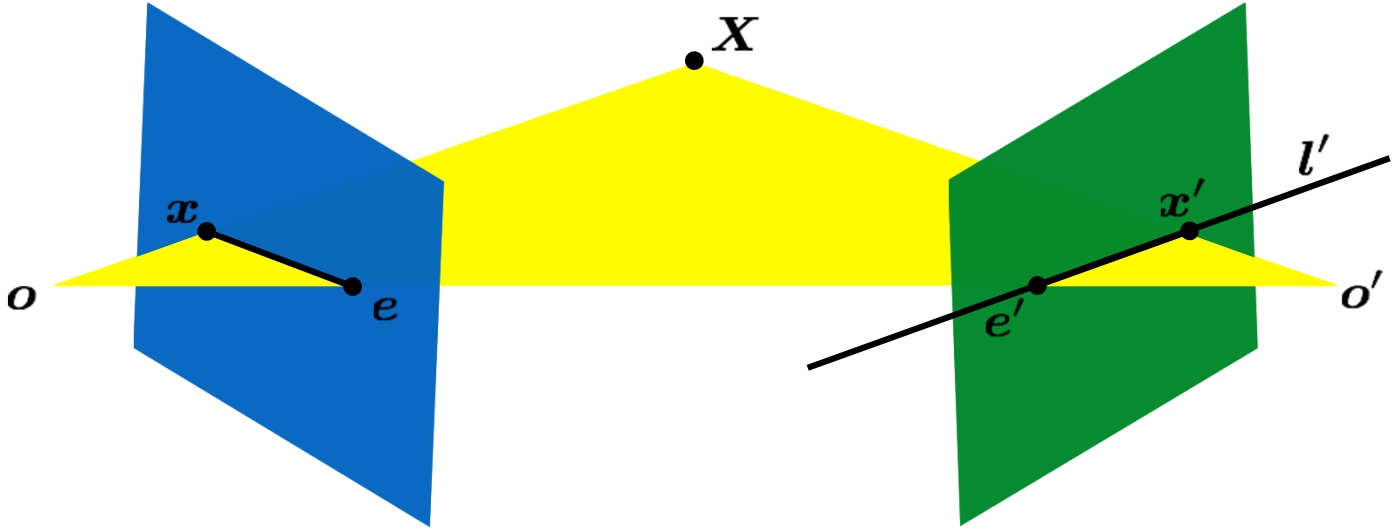
Essential Matrix

- Epipolar Line

So if $\boldsymbol{x}'^{\top}\boldsymbol{l}' = 0$ and $\mathbf{E}\boldsymbol{x} = \boldsymbol{l}'$ then

$\boldsymbol{x}'^{\top}\mathbf{E}\boldsymbol{x} = \ ?$

Essential Matrix
[Longuet-Higgins 1981]



Essential Matrix

- properties of the E matrix

  - Longuet-Higgins equation $\boldsymbol{x}'^{\top}\mathbf{E}\boldsymbol{x}=0$

  - Epipolar lines $\boldsymbol{x}^{\top}\boldsymbol{l}=0$ $\quad \boldsymbol{x}'^{\top}\boldsymbol{l}'=0$

    $\boldsymbol{l}'=\mathbf{E}\boldsymbol{x}$ $\quad \boldsymbol{l}=\mathbf{E}^{T}\boldsymbol{x}'$

  - Epipoles $\boldsymbol{e}'^{\top}\mathbf{E}=\mathbf{0}$ $\quad \mathbf{E}\boldsymbol{e}=\mathbf{0}$

(2D points expressed in camera coordinate system)

Essential Matrix

- Essential Matrix vs Homography
  - What's the difference between the **essential matrix** and a **homography**?

They are both 3 x 3 matrices but …

$$l' = \mathbf{E}x$$

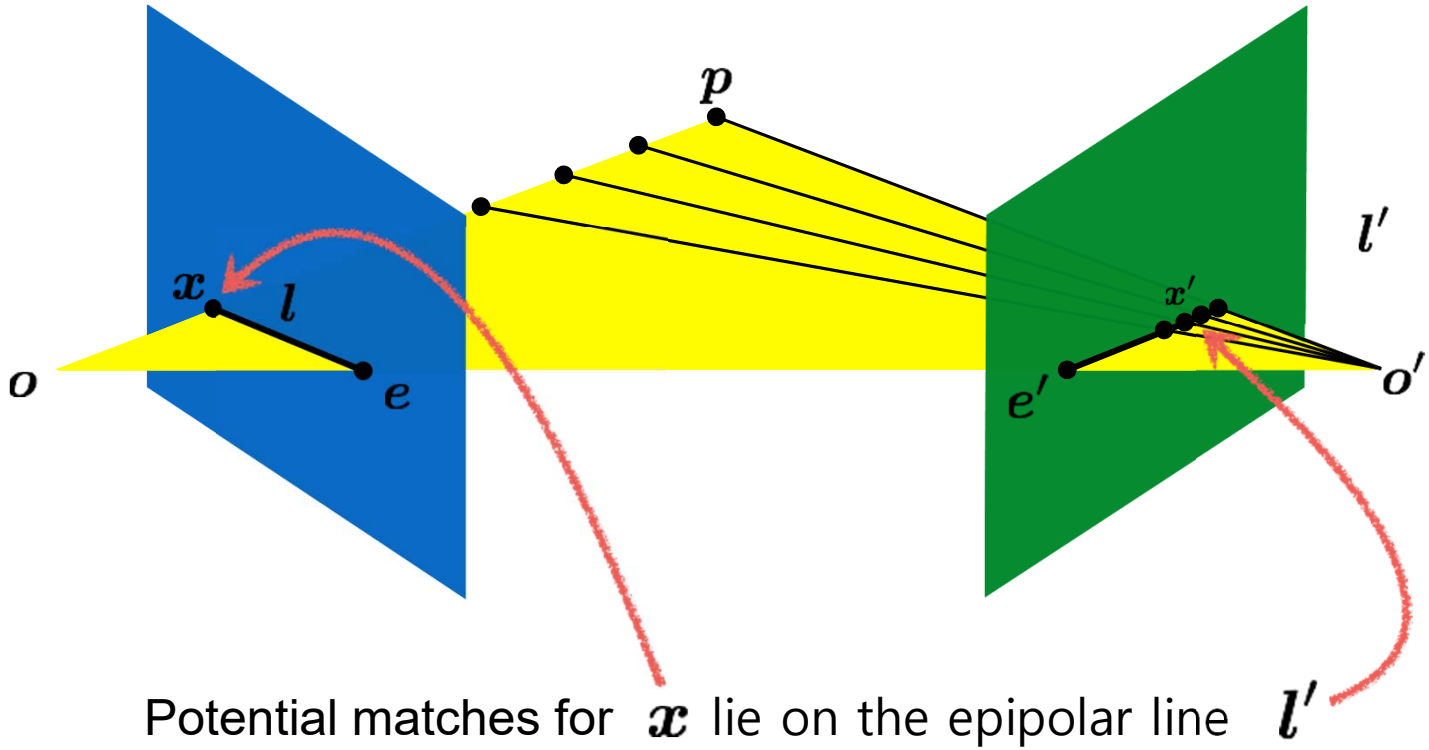
Essential matrix maps a **point** to a **line**

$$x' = \mathbf{H}x$$

Homography maps a **point** to a **point**

Essential Matrix

- **Assumption** of Essential Matrix
  - points aligned to camera coordinate axis (calibrated camera)



Potential matches for x lie on the epipolar line l'

Fundamental Matrix

- How do you generalize to non-identity intrinsic matrices?
- The **Fundamental matrix** is a **generalization** of the **Essential matrix**

  ,where the assumption of **calibrated cameras** is removed

$$\hat{\boldsymbol{x}}'^{\top}\mathbf{E}\hat{\boldsymbol{x}} = 0$$

The essential matrix operates on image points expressed in
2D coordinates expressed in the camera coordinate system

$$\hat{\boldsymbol{x}}' = \mathbf{K}'^{-1}\boldsymbol{x}'$$

$$\hat{\boldsymbol{x}} = \mathbf{K}^{-1}\boldsymbol{x}$$

camera point

image point

Fundamental Matrix

- Writing out the epipolar constraint in terms of image coordinates

$$\boldsymbol{x}'^{\top}\mathbf{K}'^{-\top}\mathbf{E}\mathbf{K}^{-1}\boldsymbol{x} = 0$$

$$\boldsymbol{x}'^{\top}(\mathbf{K}'^{-\top}\mathbf{E}\mathbf{K}^{-1})\boldsymbol{x} = 0$$

$$\boldsymbol{x}'^{\top}\mathbf{F}\boldsymbol{x} = 0$$

Fundamental Matrix
it maps pixels to epipolar lines

Fundamental Matrix

- properties of the F matrix

  - Longuet-Higgins equation $\boldsymbol{x}'^{\top}\mathbf{F}\boldsymbol{x}=0$

  - Epipolar lines $\boldsymbol{x}^{\top}\boldsymbol{l}=0$ $\quad$ $\boldsymbol{x}'^{\top}\boldsymbol{l}'=0$

    $\boldsymbol{l}'=\mathbf{F}\boldsymbol{x}$ $\quad$ $\boldsymbol{l}=\mathbf{F}^{T}\boldsymbol{x}'$

  - Epipoles $\boldsymbol{e}'^{\top}\mathbf{F}=\mathbf{0}$ $\quad$ $\mathbf{F}\boldsymbol{e}=\mathbf{0}$

(points in image coordinates)

Fundamental Matrix

- Decompose the Fundamental Matrix
  - Breaking down the fundamental matrix

$$\mathbf{F} = \mathbf{K}'^{-\top}\mathbf{E}\mathbf{K}^{-1}$$

$$\mathbf{F} = \mathbf{K}'^{-\top}[\mathbf{t}_\times]\mathbf{R}\mathbf{K}^{-1}$$

  - Depends on both intrinsic and extrinsic parameters

$$\boldsymbol{x}'^{\top}_m \mathbf{F} \boldsymbol{x}_m = 0$$

How would you solve for F?

Eight-Point Algorithm

- Assume you have *M* matched *image* points

$$\{\boldsymbol{x}_m, \boldsymbol{x}'_m\} \qquad m = 1, \ldots, M$$

- Each correspondence should satisfy

$$\boldsymbol{x}'^{\top}_m \mathbf{F} \boldsymbol{x}_m = 0$$

- *How would you solve for the 3 x 3 **F** matrix?*
  - Set up a homogeneous linear system with 9 unknowns

Eight-Point Algorithm

- How many equation do you get from one correspondence?

$$\boldsymbol{x}_m'^{\top}\mathbf{F}\boldsymbol{x}_m = 0$$

$$\begin{bmatrix} x_m' & y_m' & 1 \end{bmatrix} \begin{bmatrix} f_1 & f_2 & f_3 \\ f_4 & f_5 & f_6 \\ f_7 & f_8 & f_9 \end{bmatrix} \begin{bmatrix} x_m \\ y_m \\ 1 \end{bmatrix} = 0$$

  - ONE correspondence gives you ONE equation

$$\begin{aligned} x_m x_m' f_1 + x_m y_m' f_2 + x_m f_3 + \\ y_m x_m' f_4 + y_m y_m' f_5 + y_m f_6 + \\ x_m' f_7 + y_m' f_8 + f_9 = 0 \end{aligned}$$

Eight-Point Algorithm

- How many equation do you get from one correspondence?

$$\boldsymbol{x}_m'^{\top}\mathbf{F}\boldsymbol{x}_m = 0$$

$$\begin{bmatrix} x_m' & y_m' & 1 \end{bmatrix} \begin{bmatrix} f_1 & f_2 & f_3 \\ f_4 & f_5 & f_6 \\ f_7 & f_8 & f_9 \end{bmatrix} \begin{bmatrix} x_m \\ y_m \\ 1 \end{bmatrix} = 0$$

  - Set up a homogeneous linear system with 9 unknowns

$$\begin{bmatrix} x_1x_1' & x_1y_1' & x_1 & y_1x_1' & y_1y_1' & y_1 & x_1' & y_1' & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_Mx_M' & x_My_M' & x_M & y_Mx_M' & y_My_M' & y_M & x_M' & y_M' & 1 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \\ f_6 \\ f_7 \\ f_8 \\ f_9 \end{bmatrix} = \mathbf{0}$$

Eight-Point Algorithm

- How do you solve a homogeneous linear system?

$$\mathbf{A}\boldsymbol{X} = \mathbf{0}$$

- Total Least Squares

minimize $\|\mathbf{A}\boldsymbol{x}\|^2$

subject to $\|\boldsymbol{x}\|^2 = 1$

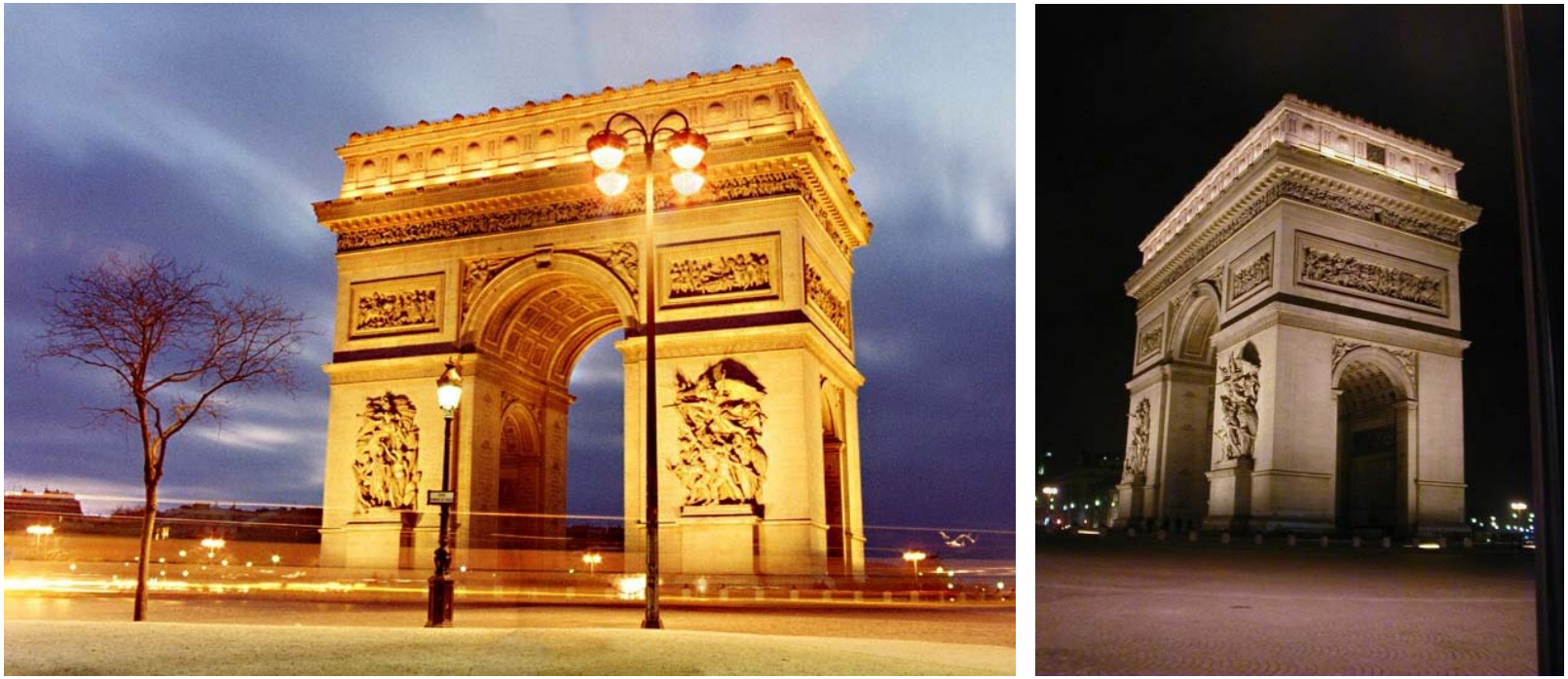
Eight-Point Algorithm

0. (Normalize points)
1. Construct the M x 9 matrix **A**
2. Find the SVD of **A**
3. Entries of **F** are the elements of column of **V** corresponding to the least singular value
4. (Enforce rank 2 constraint on F)
5. (Un-normalize F)

Fundamental Matrix

- Example

$$\mathbf{F} = \begin{bmatrix} -0.00310695 & -0.0025646 & 2.96584 \\ -0.028094 & -0.00771621 & 56.3813 \\ 13.1905 & -29.2007 & -9999.79 \end{bmatrix}$$



Input images

Fundamental Matrix

- Example

$$\mathbf{F} = \begin{bmatrix} -0.00310695 & -0.0025646 & 2.96584 \\ -0.028094 & -0.00771621 & 56.3813 \\ 13.1905 & -29.2007 & -9999.79 \end{bmatrix}$$



$$\boldsymbol{x} = \begin{bmatrix} 343.53 \\ 221.70 \\ 1.0 \end{bmatrix}$$

$$\boldsymbol{l}' = \mathbf{F}\boldsymbol{x} = \begin{bmatrix} 0.0295 \\ 0.9996 \\ -265.1531 \end{bmatrix}$$

Calculating epipolar lines

Fundamental Matrix

- Example

$$l' = \mathbf{F}x = \begin{bmatrix} 0.0295 \\ 0.9996 \\ -265.1531 \end{bmatrix}$$



Calculating epipolar lines

Fundamental Matrix

https://docs.opencv.org/master/da/de9/tutorial_py_epipolar_geometry.html

- Example



epipolar lines

Fundamental Matrix

- Example



epipolar lines

Where is the epipole?

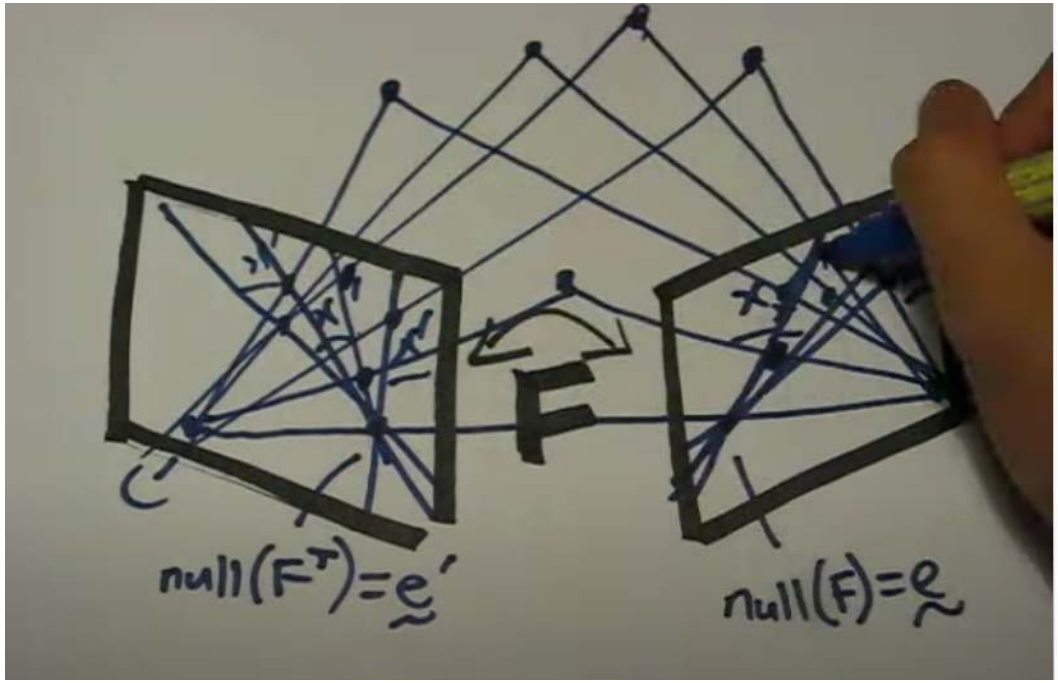
How would you compute it?

$$\mathbf{F}e = 0$$

How would you solve for the epipole?

Fundamental Matrix Song

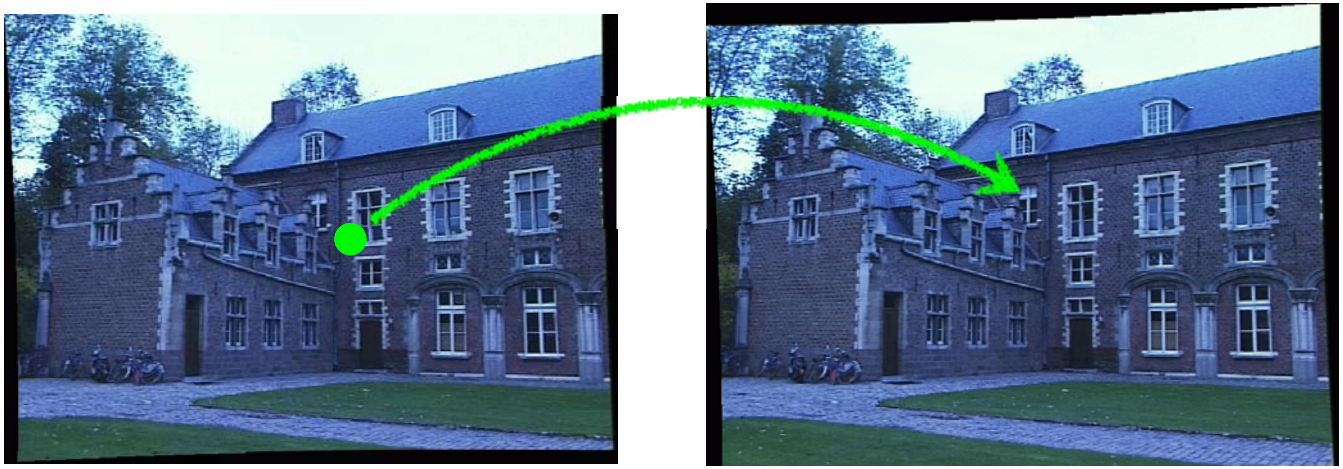
- https://www.youtube.com/watch?v=DgGV3l82NTk



Stereo Vision

Triangulation

- Task: Match point in left image to point in right image



Left image

Right image

Triangulation



Left image



Right image

Triangulation



Left image



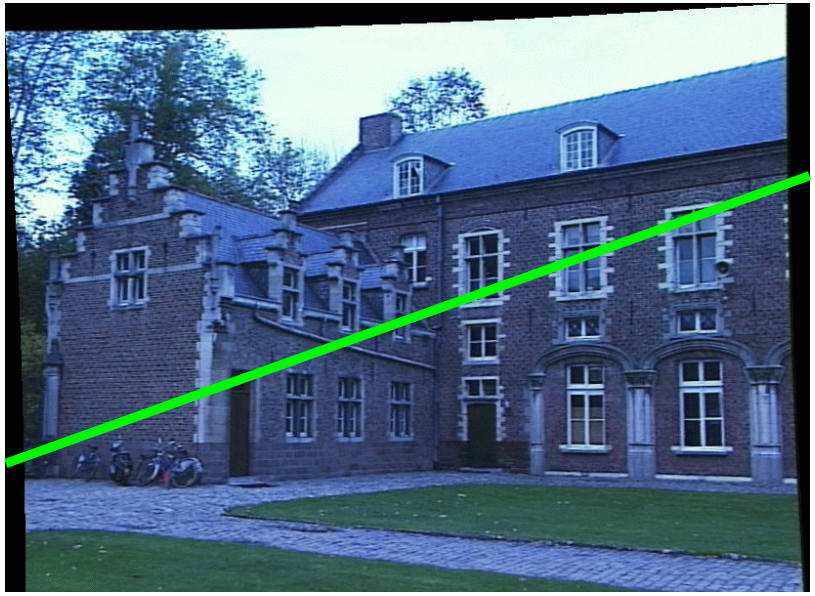
Right image

1. Select point in one image

Triangulation



Left image



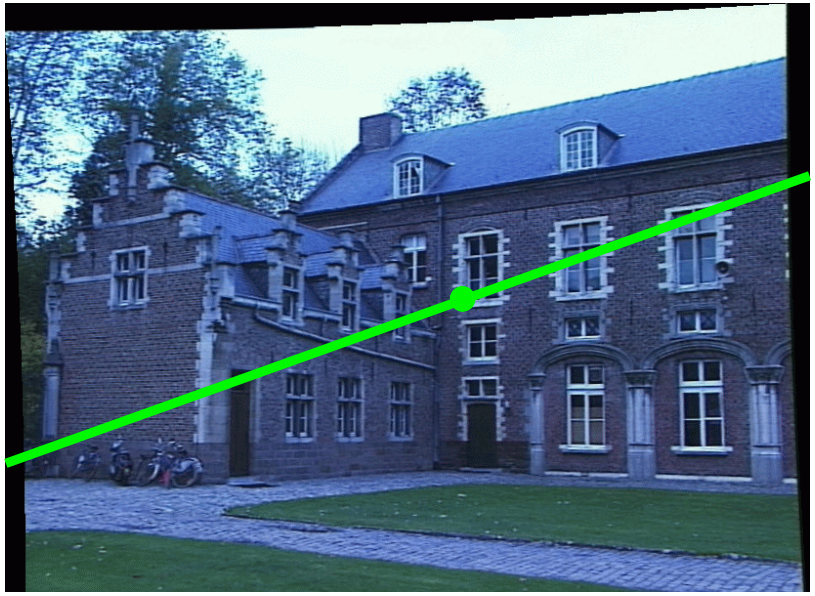
Right image

1. Select point in one image
2. Form epipolar line for that point in second image

Triangulation



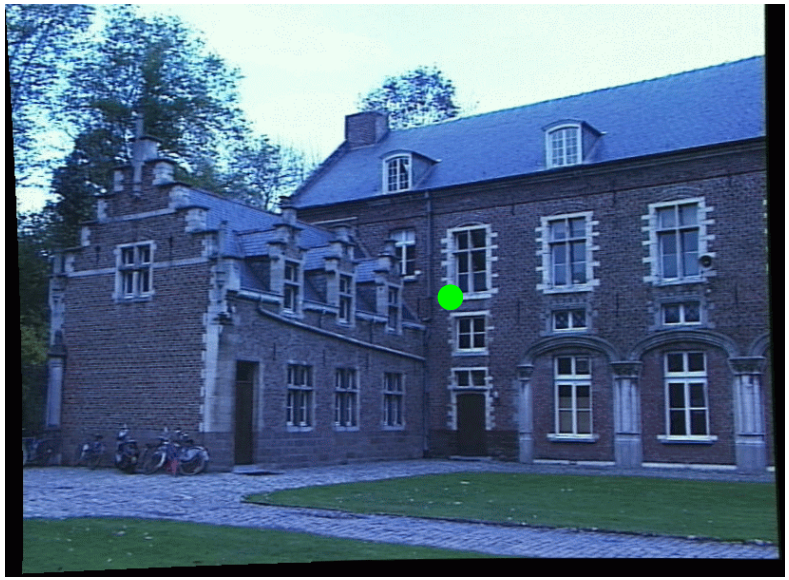
Left image



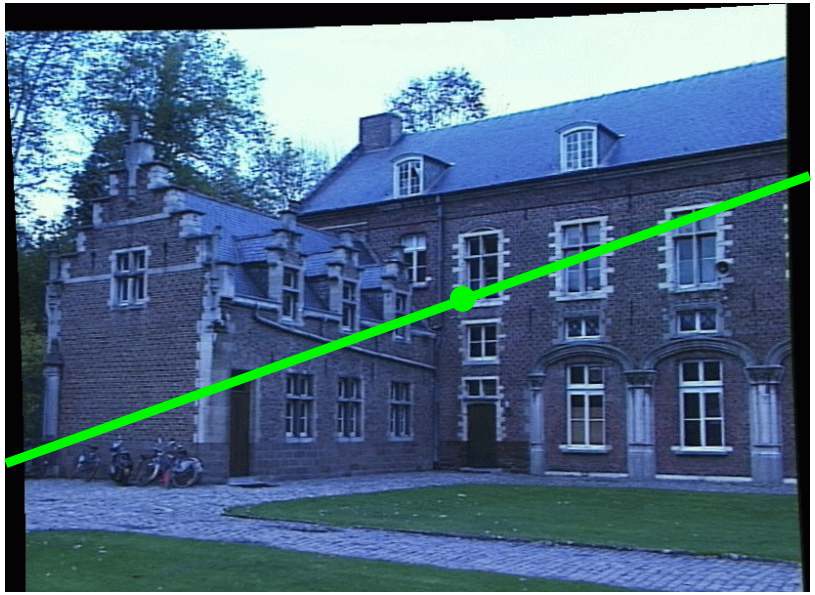
Right image

1. Select point in one image
2. Form epipolar line for that point in second image
3. Find matching point along line

Triangulation



Left image

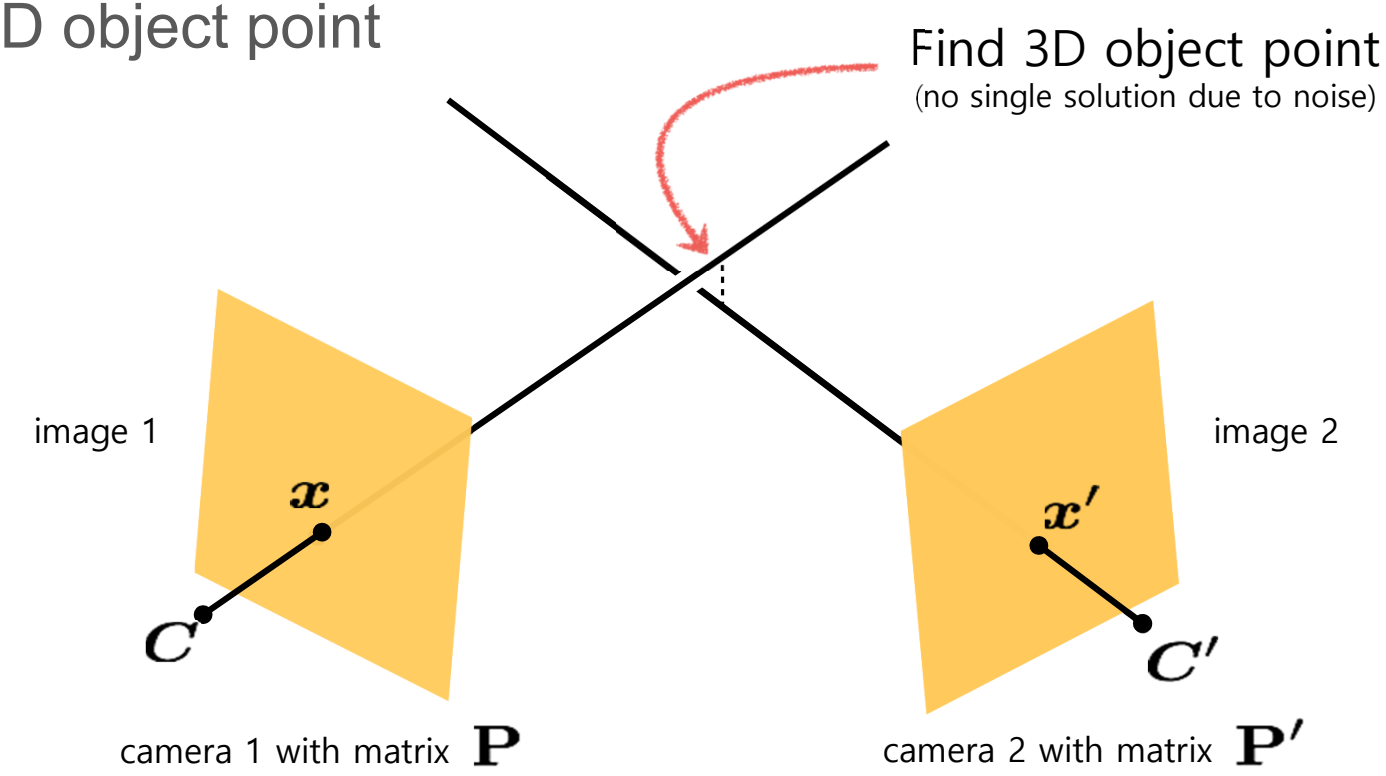


Right image

1. Select point in one image
2. Form epipolar line for that point in second image
3. Find matching point along line
4. Perform triangulation

Triangulation

- Find 3D object point



Triangulation

- Find 3D object point

Given a set of (noisy) matched points

$$\{\boldsymbol{x}_i, \boldsymbol{x}_i'\}$$

and camera matrices

$$\mathbf{P}, \mathbf{P}'$$

Estimate the 3D point

$$\mathbf{X}$$

$$\mathbf{x} = \mathbf{P}\boldsymbol{X}$$

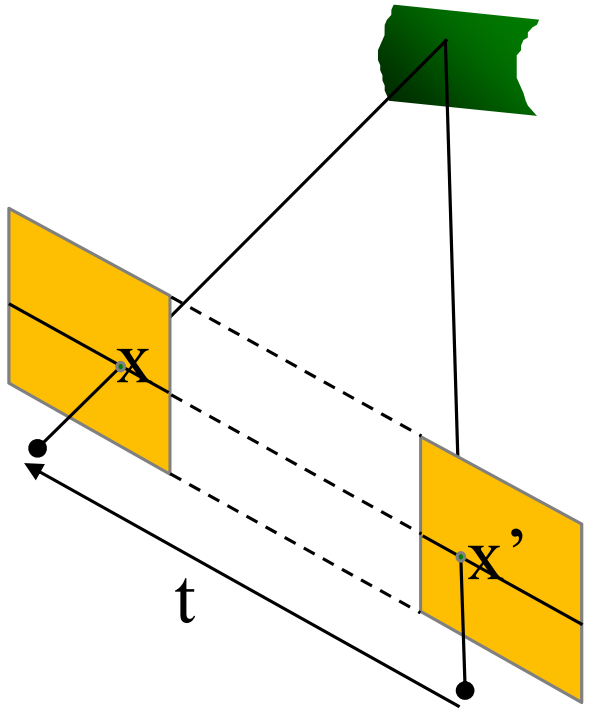
known known

Stereo rectification

- When are epipolar lines horizontal?

$$R = I \qquad t = (T, 0, 0)$$

  - Image planes of cameras are parallel to each other and to the baseline
  - Camera centers are at same height
  - Focal lengths are the same
  - Then epipolar lines fall along the horizontal scan lines of the images



Stereo rectification

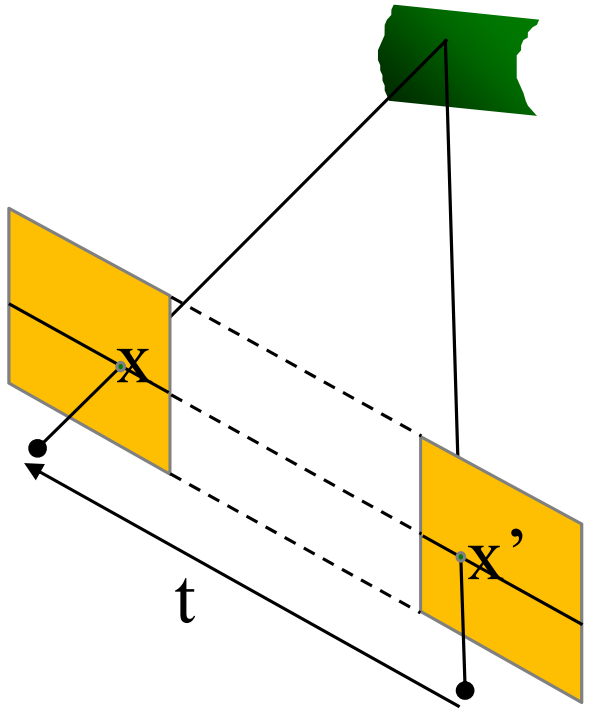
- When are epipolar lines horizontal?

$$R = I \qquad t = (T, 0, 0)$$

$$\boldsymbol{x}'^{T}\boldsymbol{E}\boldsymbol{x} = 0, \quad \boldsymbol{E} = [\boldsymbol{t}_{\times}]\boldsymbol{R}$$

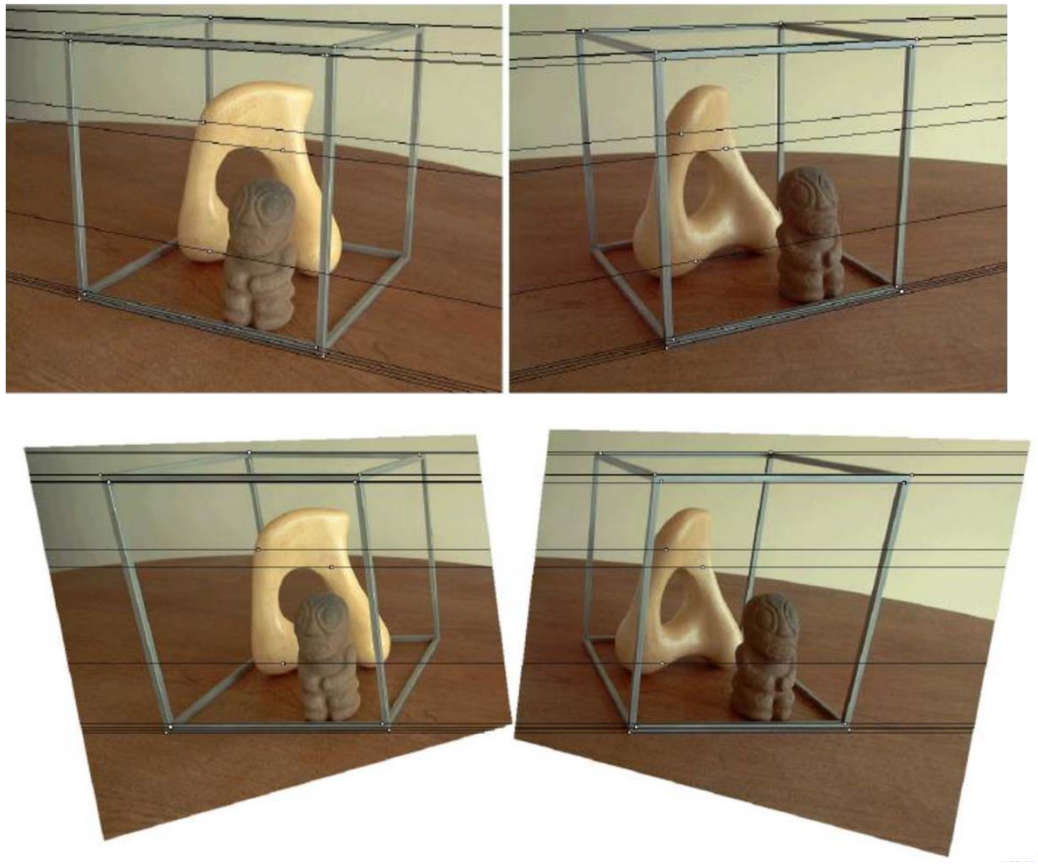
$$\boldsymbol{E} = [\boldsymbol{t}_{\times}]\boldsymbol{R} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -T \\ 0 & T & 0 \end{bmatrix}$$

$$\begin{pmatrix} u' & v' & 1 \end{pmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -T \\ 0 & T & 0 \end{bmatrix} \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = 0 \qquad \begin{pmatrix} u' & v' & 1 \end{pmatrix} \begin{pmatrix} 0 \\ -T \\ Tv \end{pmatrix} = 0 \qquad Tv' = Tv$$



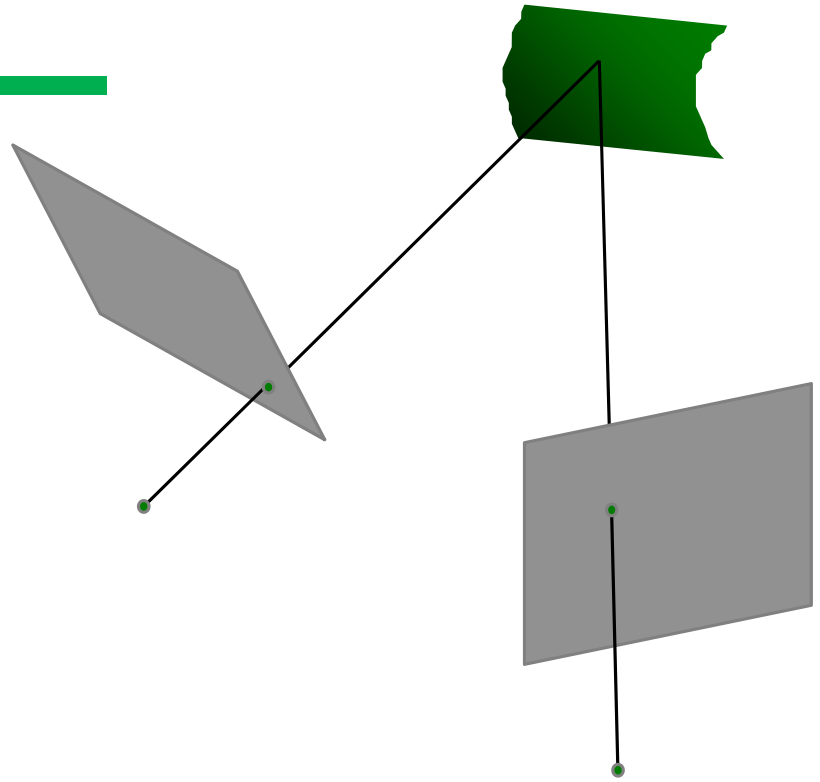
Stereo rectification

- How can you make the epipolar lines horizontal?



Stereo rectification

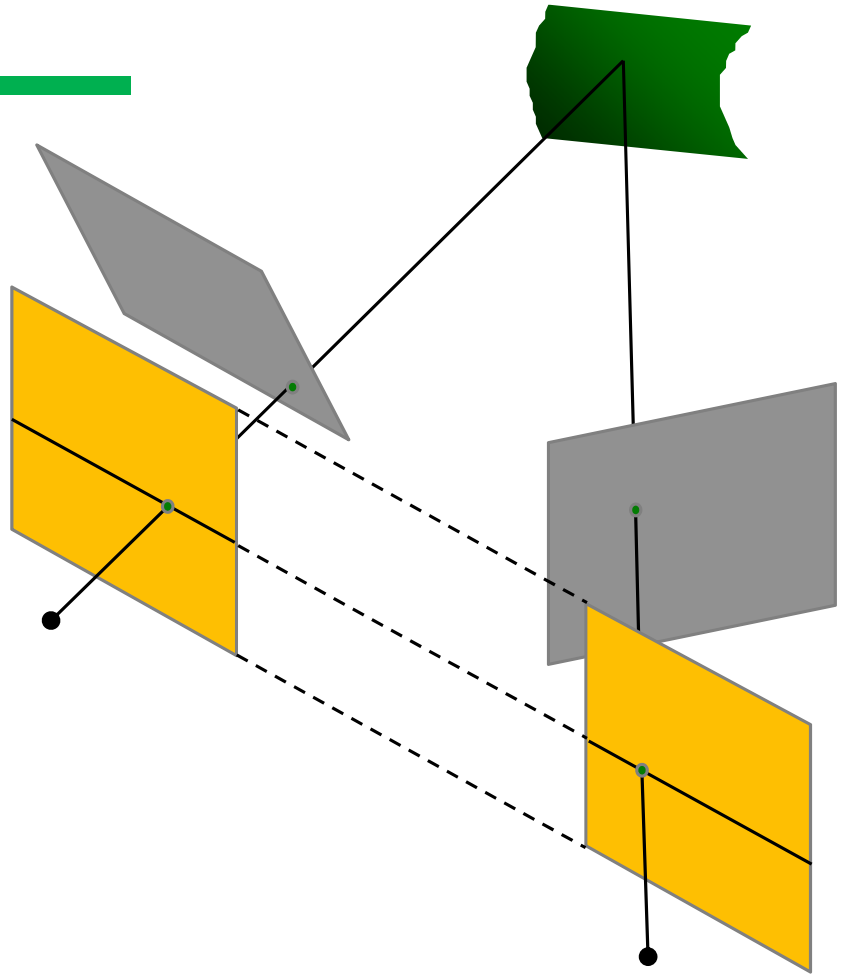
- What is stereo rectification?



C. Loop and Z. Zhang. Computing Rectifying Homographies for Stereo Vision. Computer Vision and Pattern Recognition, 1999.

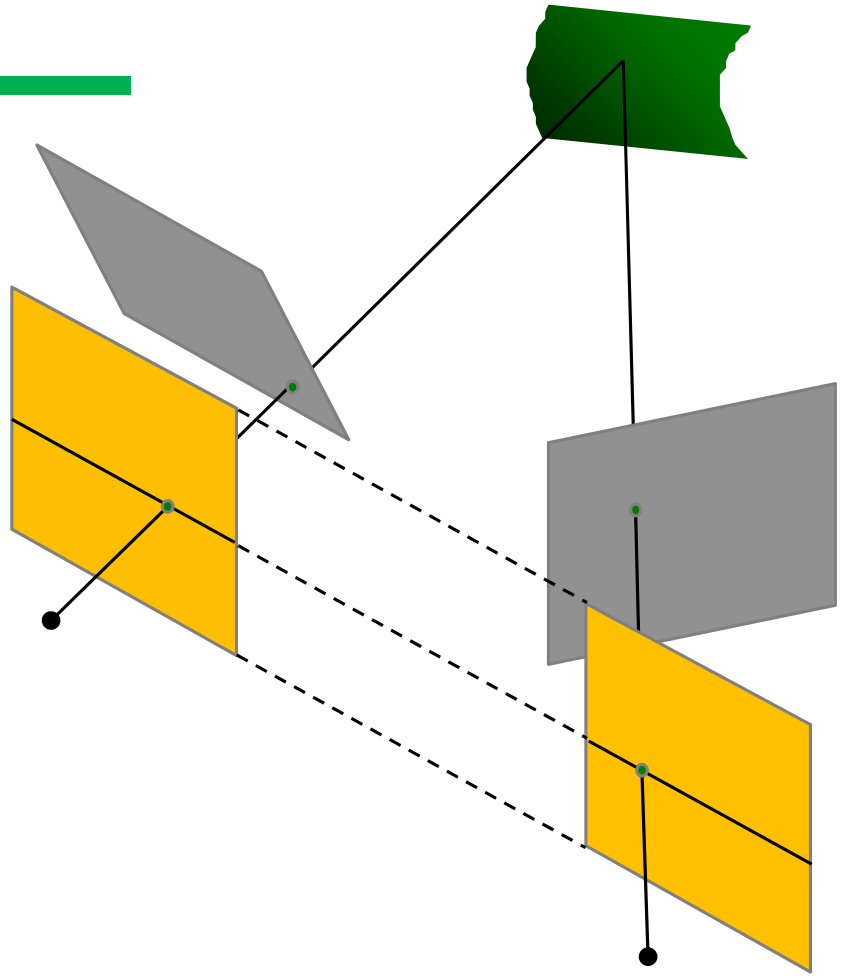
Stereo rectification

- Reproject image planes onto a common plane parallel to the line between camera centers



Stereo rectification

- Reproject image planes onto a common plane parallel to the line between camera centers
- Need two homographies (3x3 transform), one for each input image reprojection



Stereo rectification

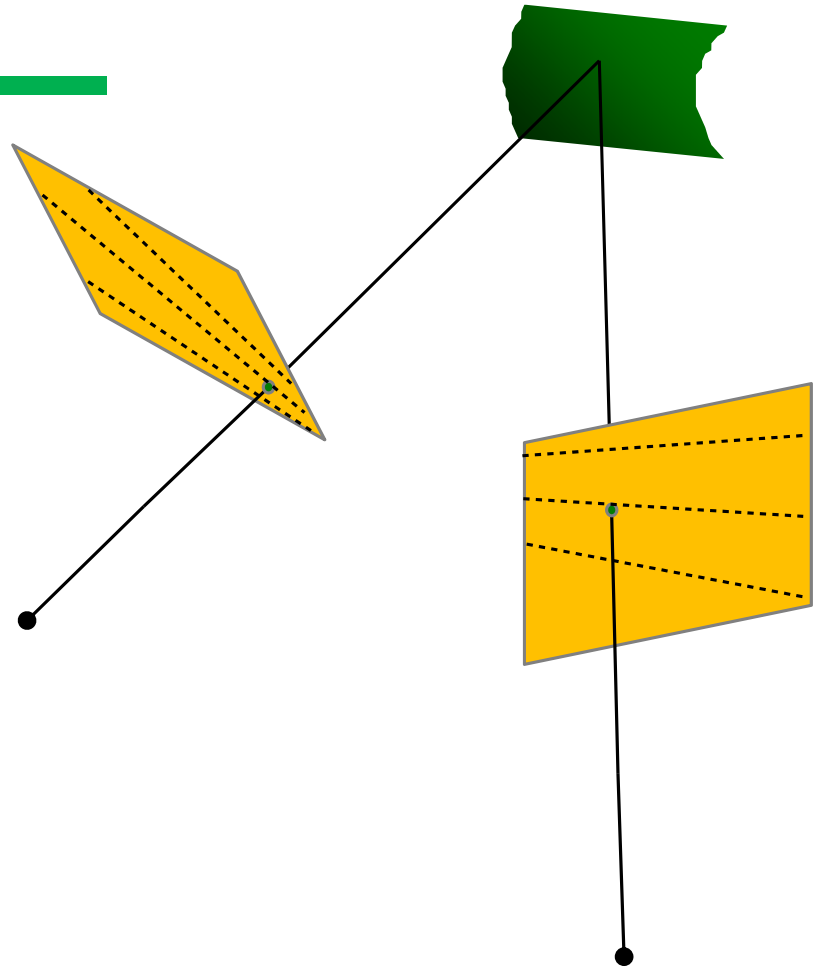
1. **Rotate** the right camera by **R**
   (aligns camera coordinate system orientation only)

2. Rotate (**rectify**) the left camera so that the epipole is at infinity

3. Rotate (**rectify**) the right camera so that the epipole is at infinity

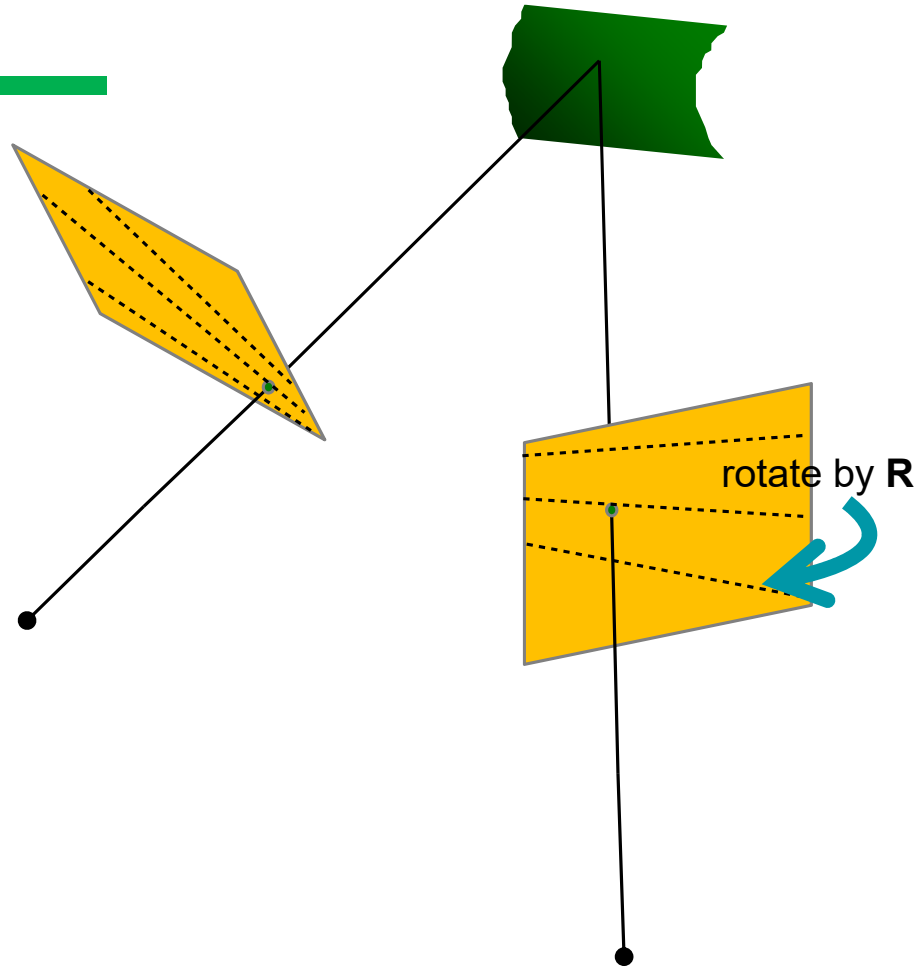
4. Adjust the **scale**

Stereo rectification



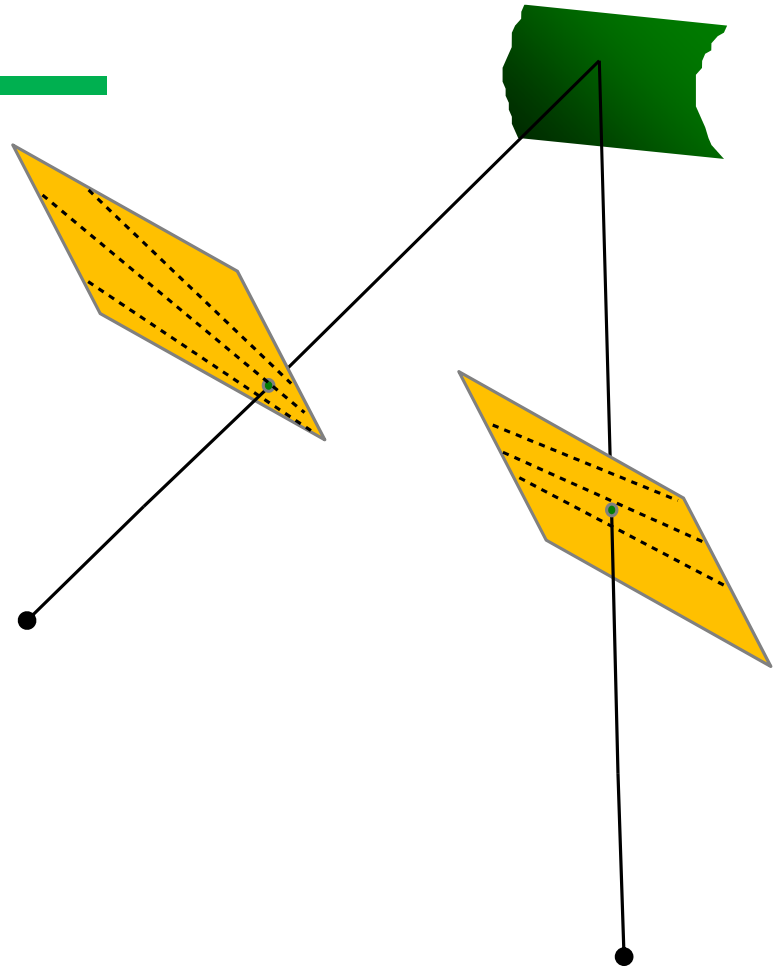
1. Compute **E** to get **R**
2. Rotate right image by **R**
3. Rotate both images by $\mathbf{R}_{rect}$
4. Scale both images by **H**

Stereo rectification



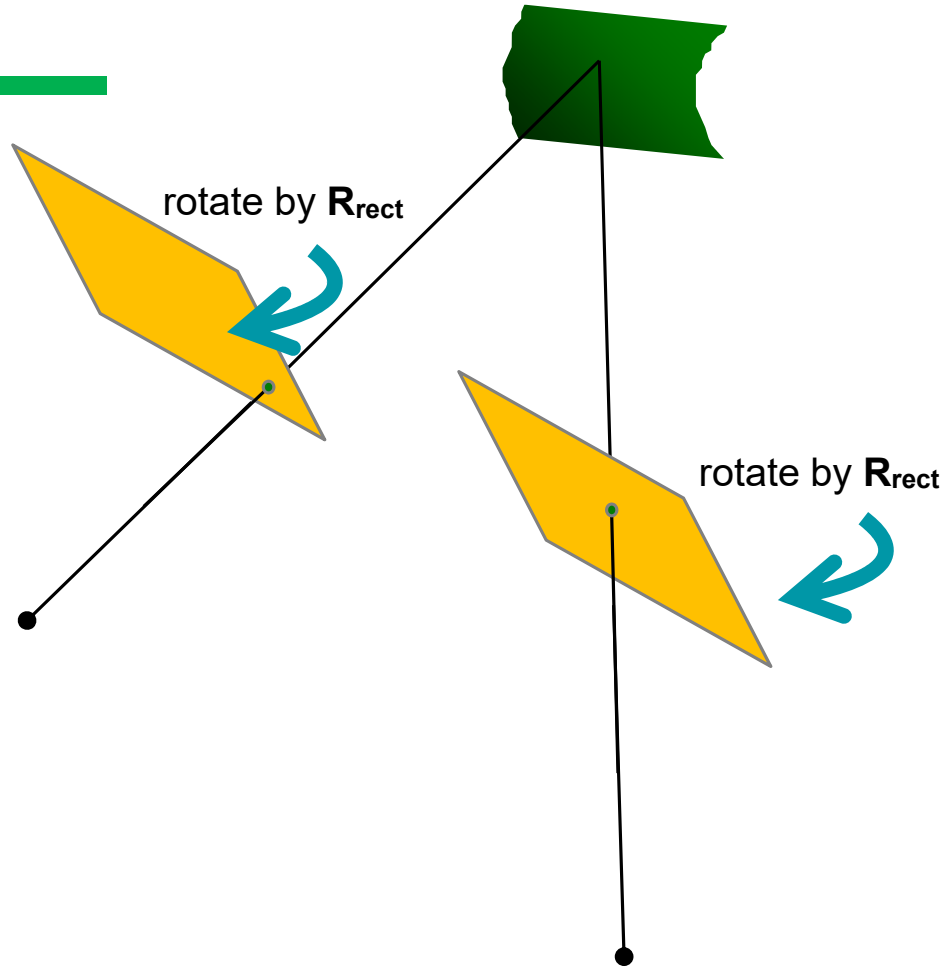
1. Compute **E** to get **R**
2. Rotate right image by **R**
3. Rotate both images by $\mathbf{R}_{rect}$
4. Scale both images by **H**

Stereo rectification



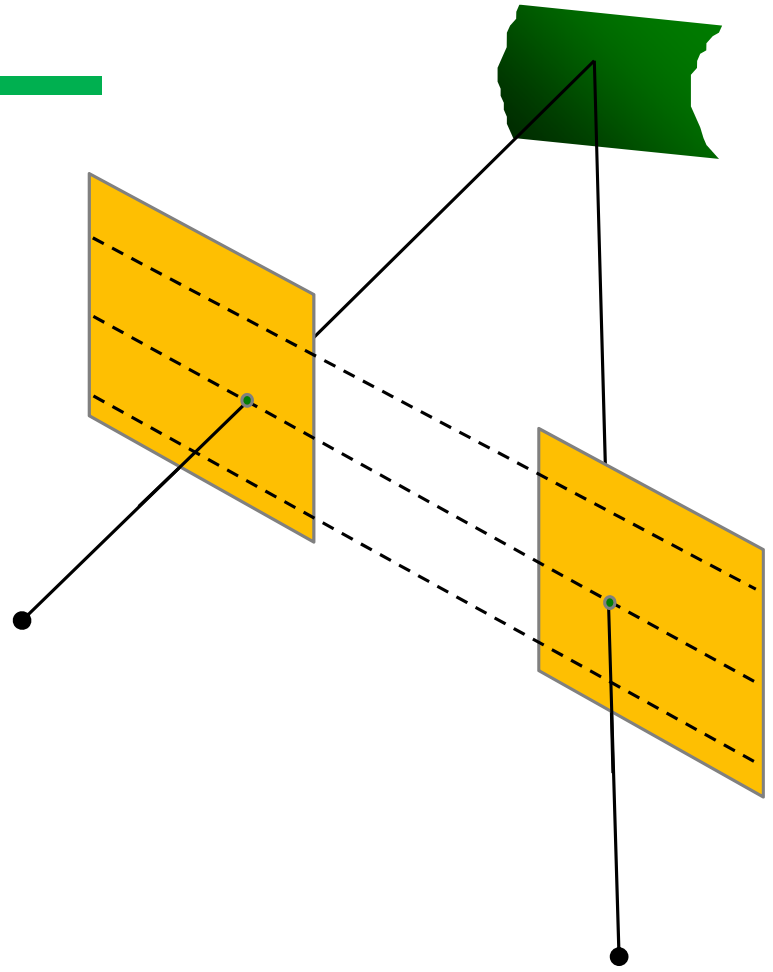
1. Compute **E** to get **R**
2. Rotate right image by **R**
3. Rotate both images by $\mathbf{R}_{rect}$
4. Scale both images by **H**

Stereo rectification



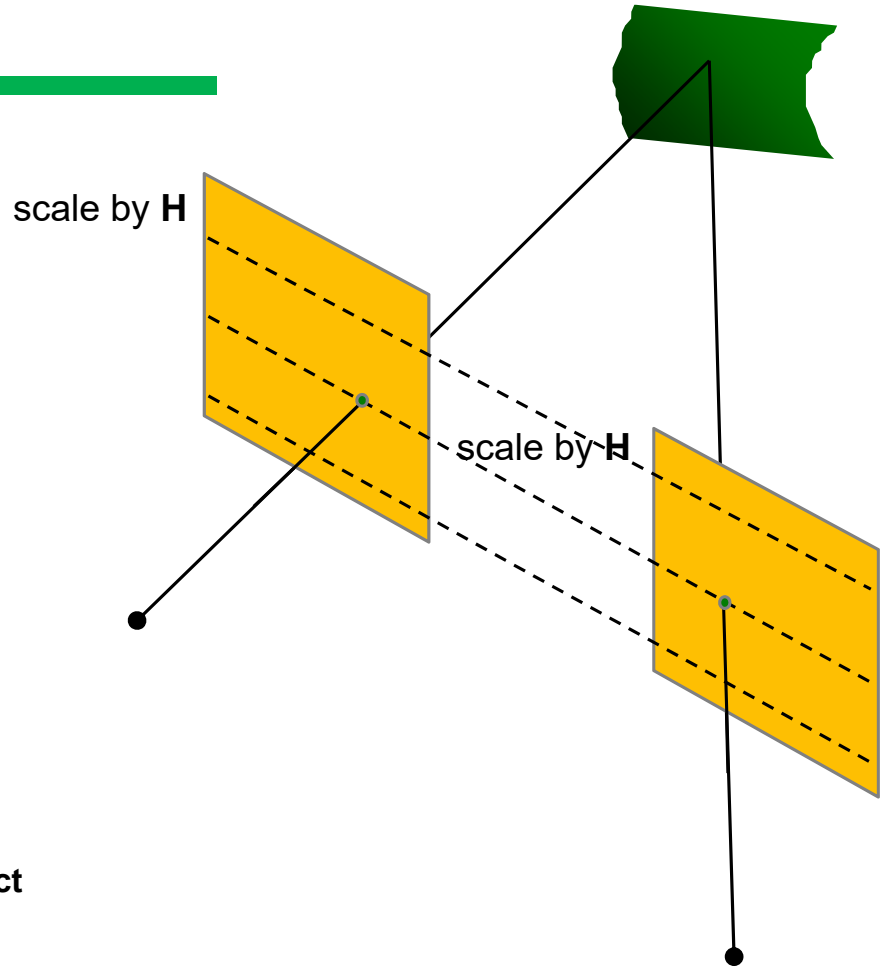
1. Compute **E** to get **R**
2. Rotate right image by **R**
3. Rotate both images by $\mathbf{R}_{rect}$
4. Scale both images by **H**

Stereo rectification



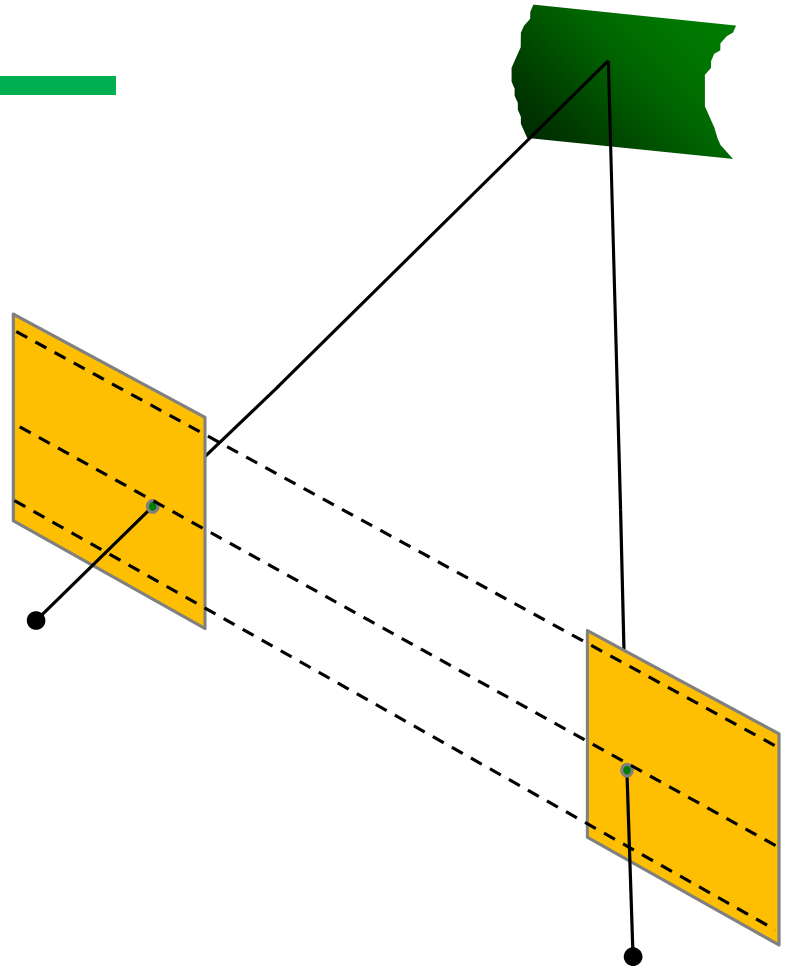
1. Compute **E** to get **R**
2. Rotate right image by **R**
3. Rotate both images by $\mathbf{R}_{\mathbf{rect}}$
4. Scale both images by **H**

Stereo rectification



1. Compute **E** to get **R**
2. Rotate right image by **R**
3. Rotate both images by $\mathbf{R}_{\mathbf{rect}}$
4. Scale both images by **H**

Stereo rectification



1. Compute **E** to get **R**
2. Rotate right image by **R**
3. Rotate both images by $\mathbf{R}_{\mathbf{rect}}$
4. Scale both images by **H**

Stereo rectification

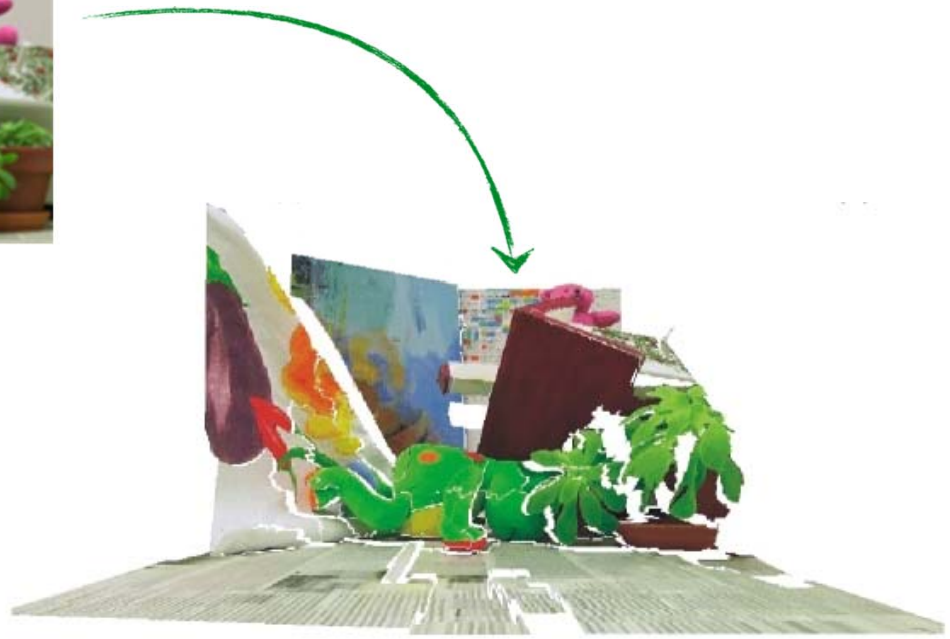
- Stereo Rectification Algorithm
  1. Estimate $\boldsymbol{E}$
  2. Estimate the epipole $\boldsymbol{e}$ (SVD of $\boldsymbol{E}$)
  3. Build $\boldsymbol{R_{rect}}$ from $\boldsymbol{e}$
  4. Decompose $\boldsymbol{E}$ into $\boldsymbol{R}$ and $\boldsymbol{T}$
  5. Set $\boldsymbol{R_1} = \boldsymbol{R_{rect}}$ and $\boldsymbol{R_2} = \boldsymbol{R R_{rect}}$
  6. Rotate each left camera point (warp image)

     $[\boldsymbol{x'}\ \boldsymbol{y'}\ \boldsymbol{z'}] = \boldsymbol{R_1}[\boldsymbol{x}\ \boldsymbol{y}\ \boldsymbol{z}]$
  7. Rectified points as $\boldsymbol{p} = \boldsymbol{f}/\boldsymbol{z'}[\boldsymbol{x'}\ \boldsymbol{y'}\ \boldsymbol{z'}]$
  8. Repeat 6 and 7 for right camera points using $\boldsymbol{R_2}$

Stereo matching

- What can we do after rectification?
- Depth Estimation via Stereo Matching

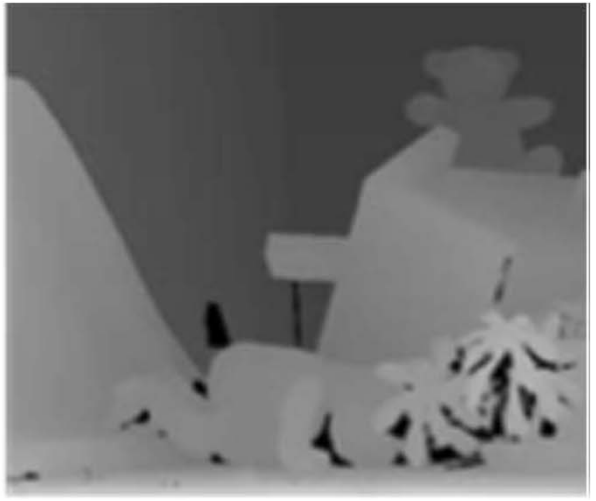




Stereo matching

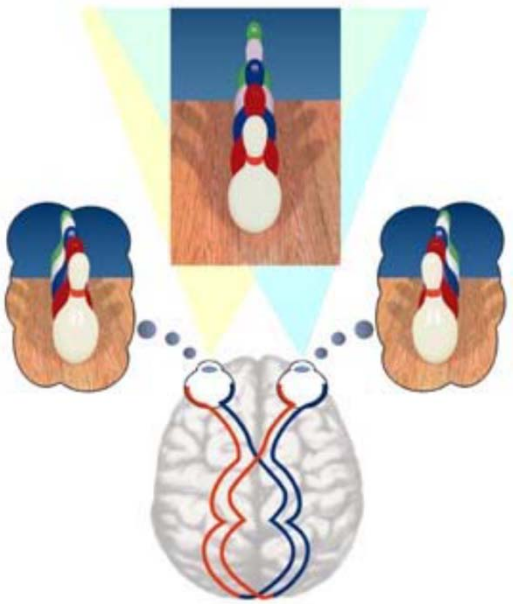
- *The amount of horizontal movement is inversely proportional to the distance from the camera*

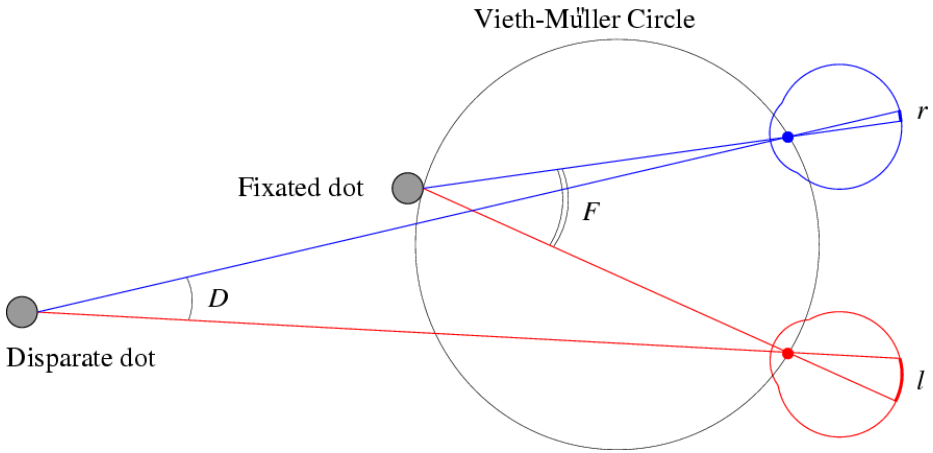




Stereo matching

- Stereopsis Concept
  - Fusing the picture recorded by our two eyes and exploiting the difference (or disparity) between them allows us to gain a strong sense of depth
  - Disparity : Occurs when eyes fixate on one object, others appear at different visual angle



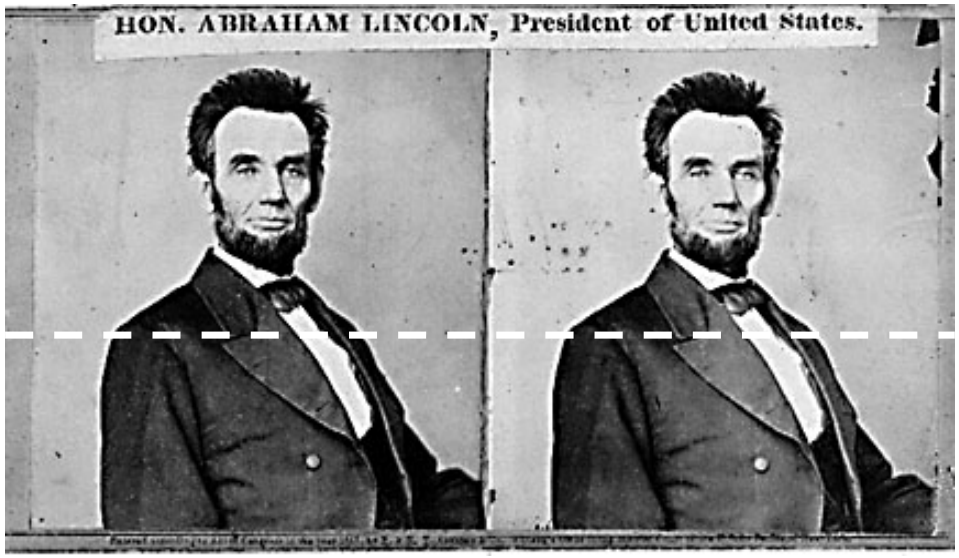


Disparity: $d = r - l = D - F$

Stereo matching

- (Basic) Stereo Matching Algorithm

  1. Rectify images (make epipolar lines horizontal)
  2. For each pixel in the first image
     - **a. Find epipolar line**
     - b. Scan line for best match
     - c. Compute depth from disparity

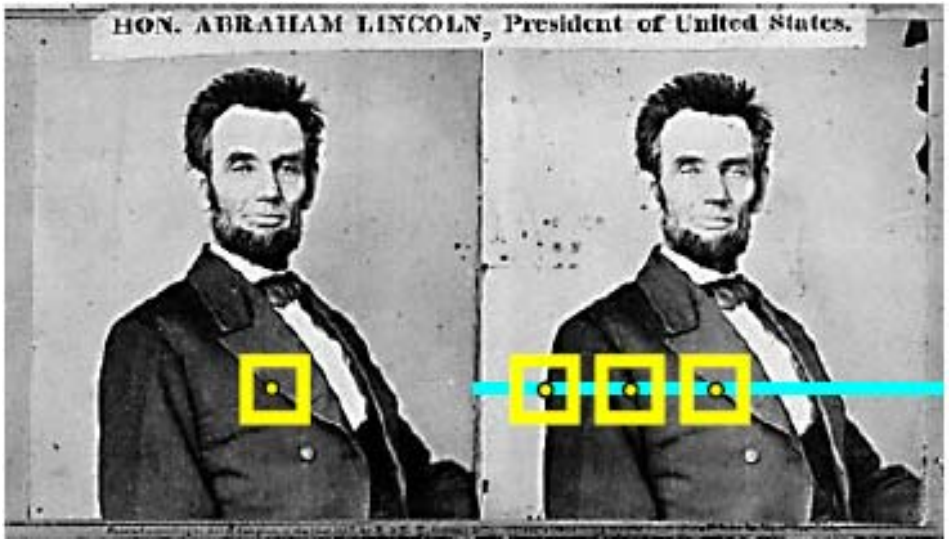


epipolar lines

Stereo matching

- (Basic) Stereo Matching Algorithm

  1. Rectify images (make epipolar lines horizontal)
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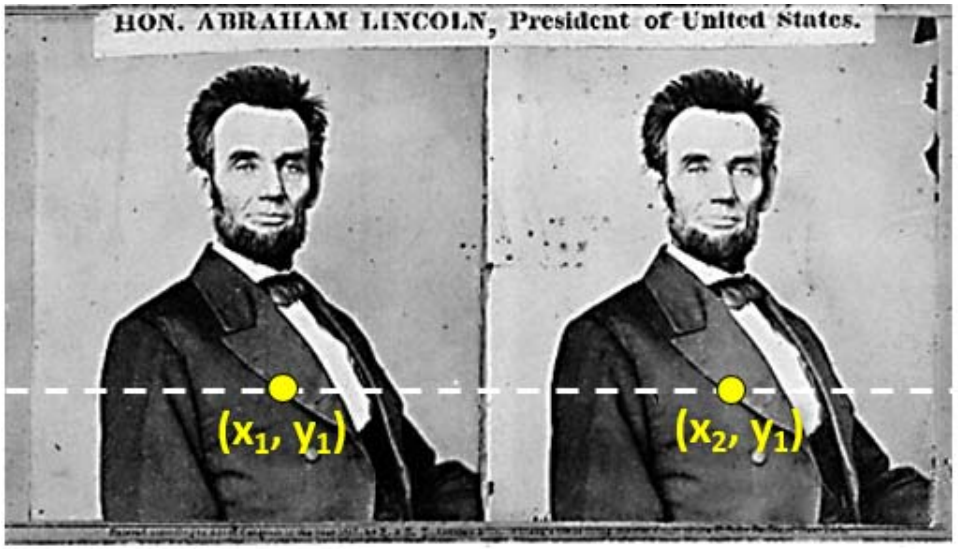


Stereo matching

- (Basic) Stereo Matching Algorithm

  1. Rectify images (make epipolar lines horizontal)
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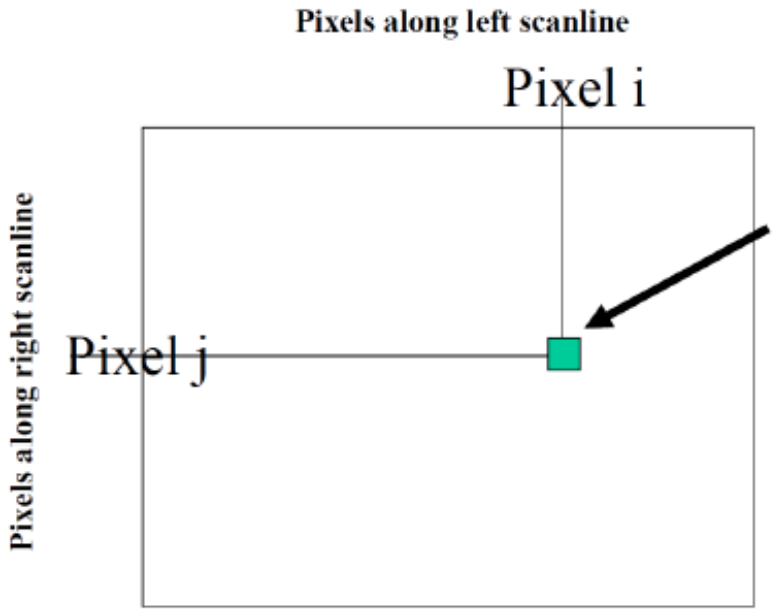
epipolar lines



$x_2 - x_1$ = the *disparity* of pixel (x_1, y_1)

Stereo matching

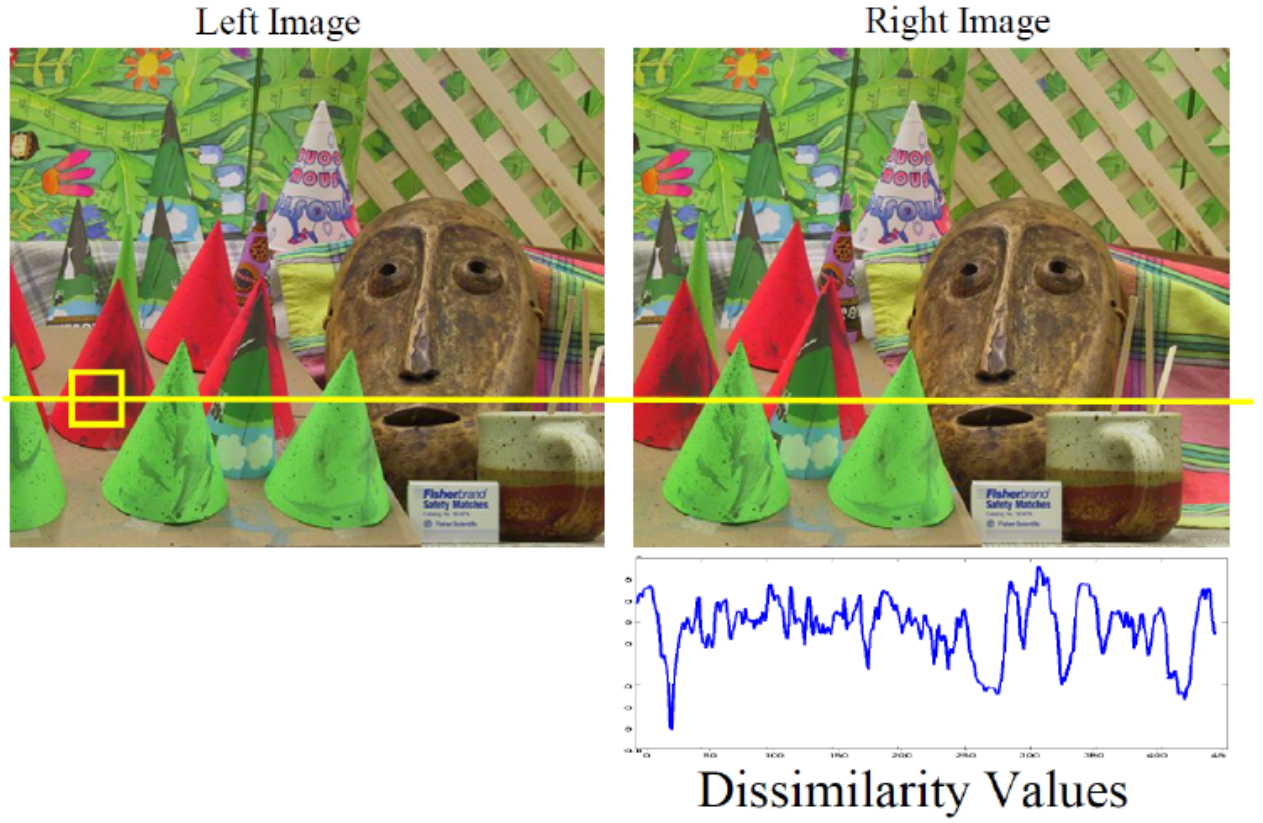
- Disparity Space Image (DSI)
  - The similarity constraint is local (each reference window is matched independently)
  - Disparity Space Image for one row



$C(i, j)$: Match score for patch centered at left pixel i with patch centered at right pixel j

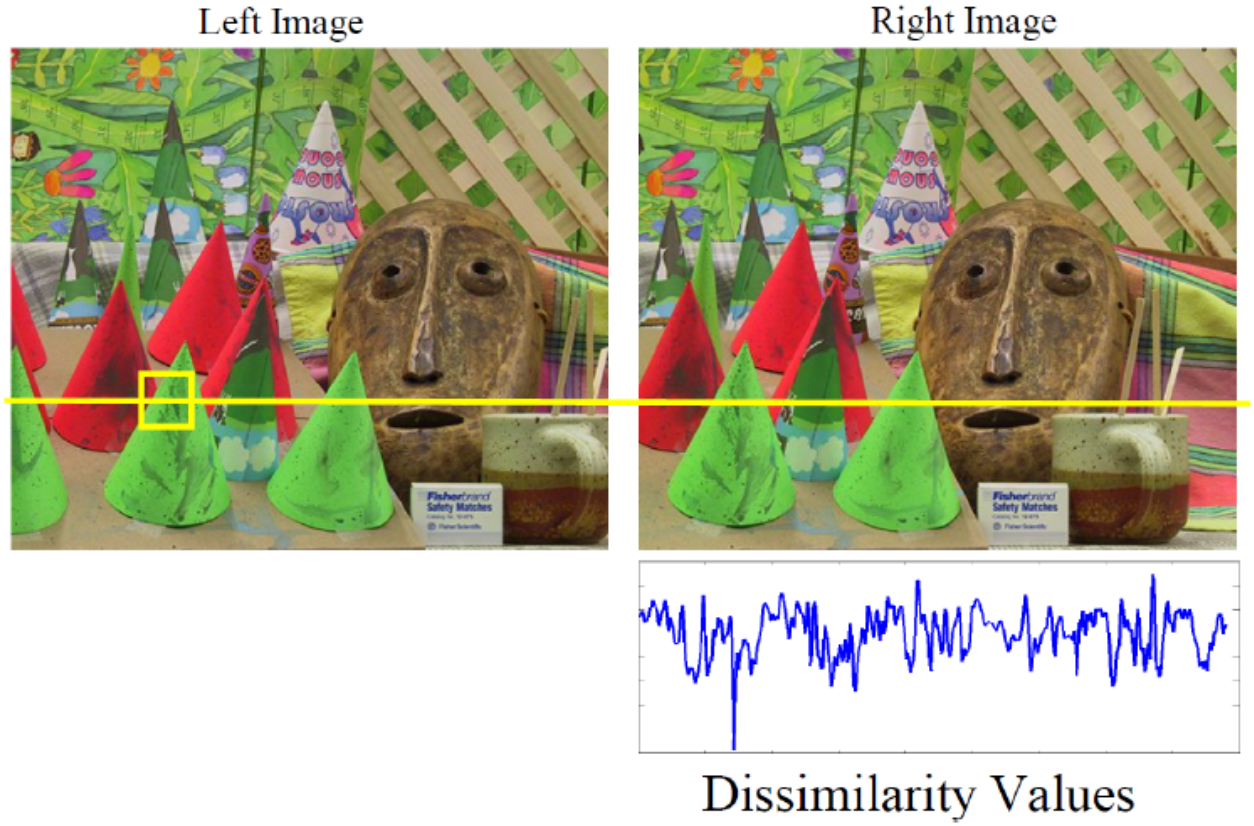
Stereo matching

- Disparity Space Image (DSI)



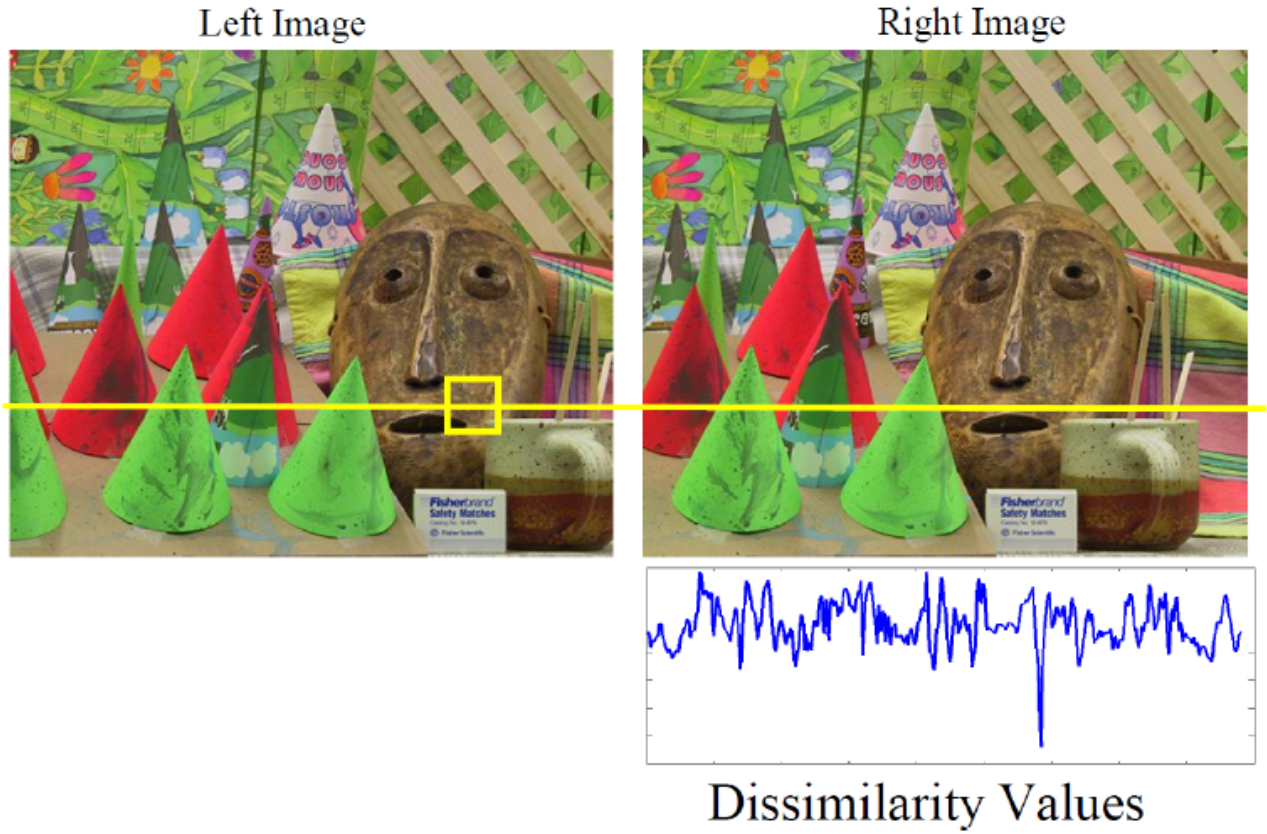
Stereo matching

- Disparity Space Image (DSI)



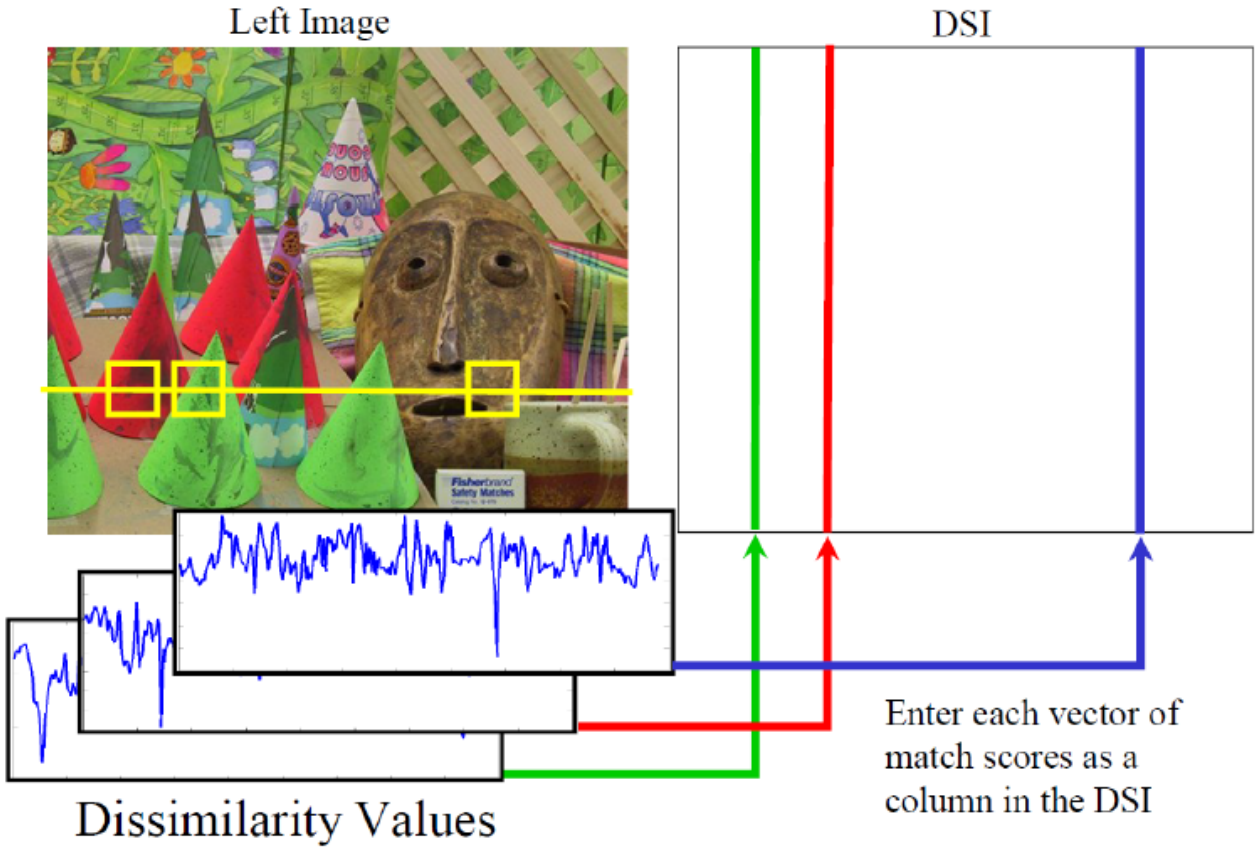
Stereo matching

- Disparity Space Image (DSI)



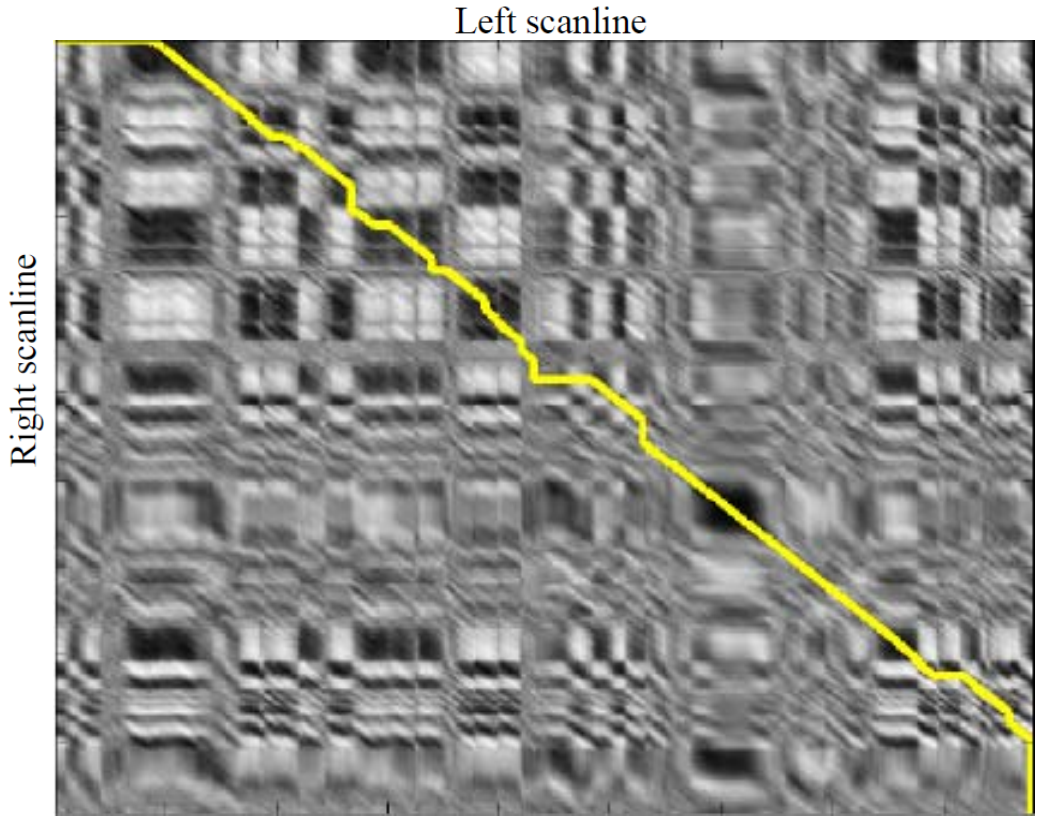
Stereo matching

- Disparity Space Image (DSI)



Stereo matching

- Disparity Space Image (DSI)



Depth from Disparity

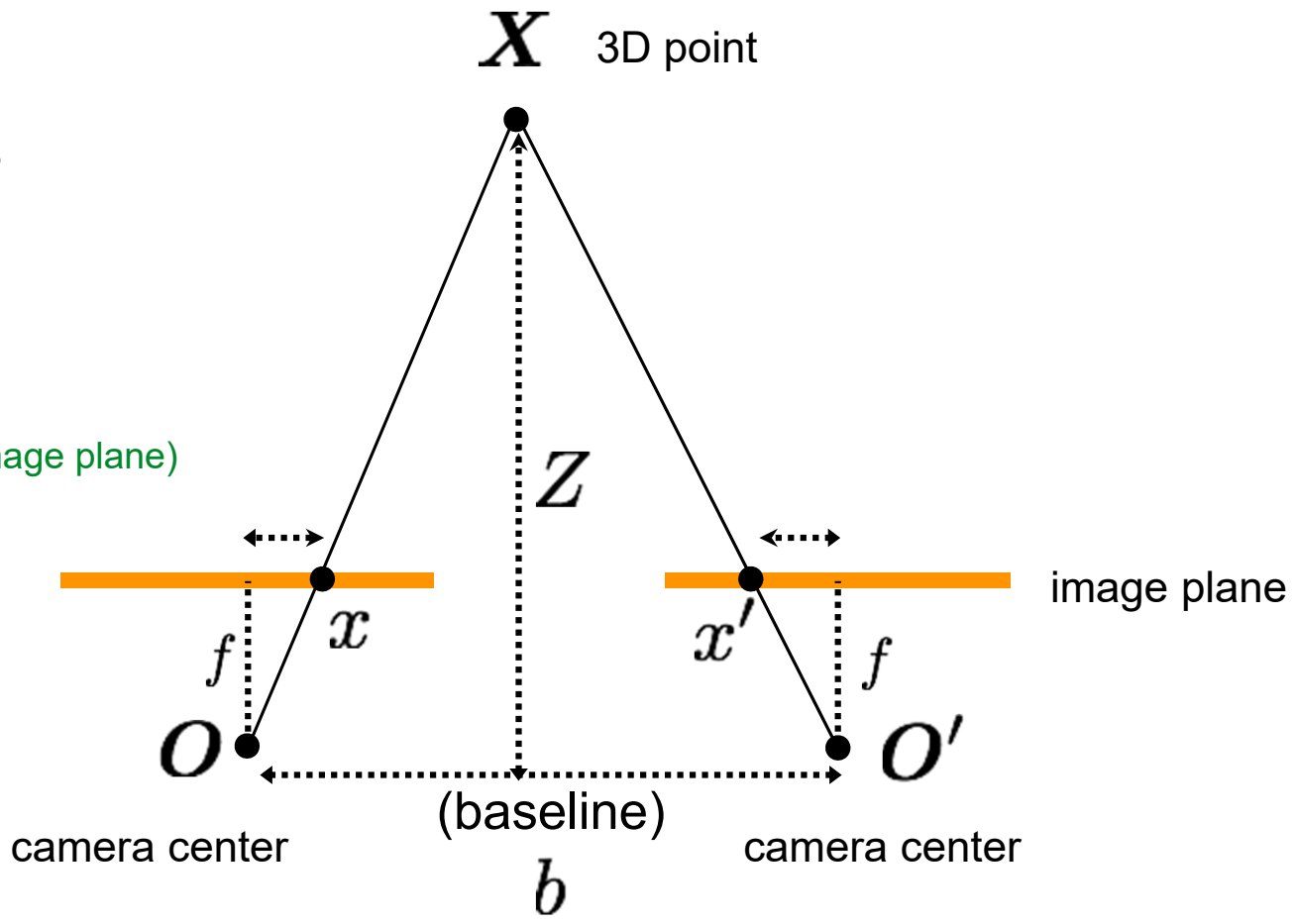
- Disparity

$$d = x - x'$$

$$= \frac{bf}{Z}$$

(wrt to camera origin of image plane)

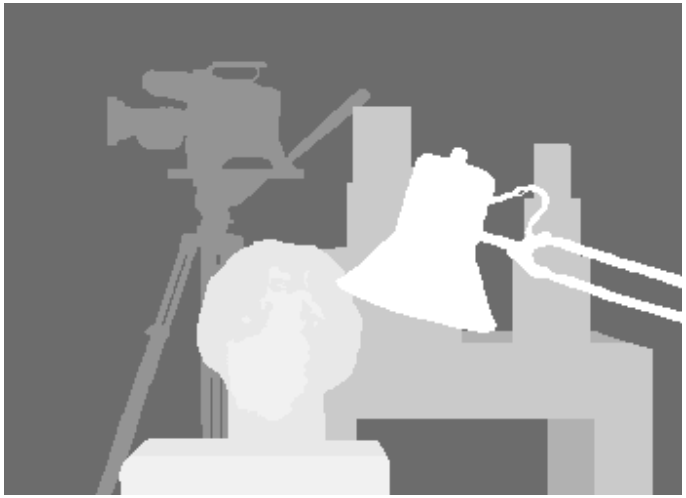
inversely proportional to depth



Depth from Disparity

- Example





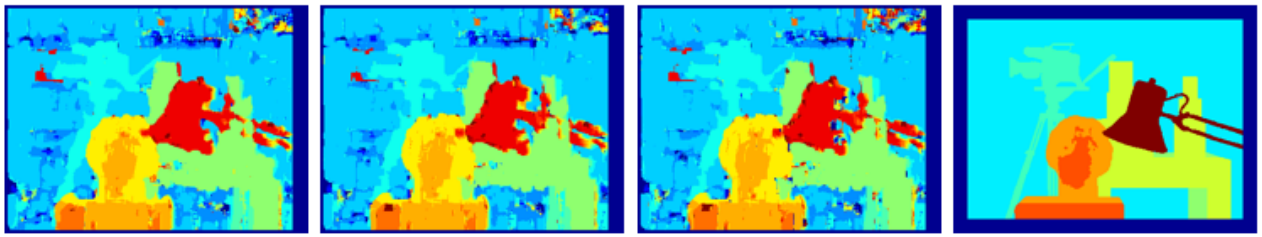
Depth from Disparity

https://docs.opencv.org/master/dd/d53/tutorial_py_depthmap.html

- Example



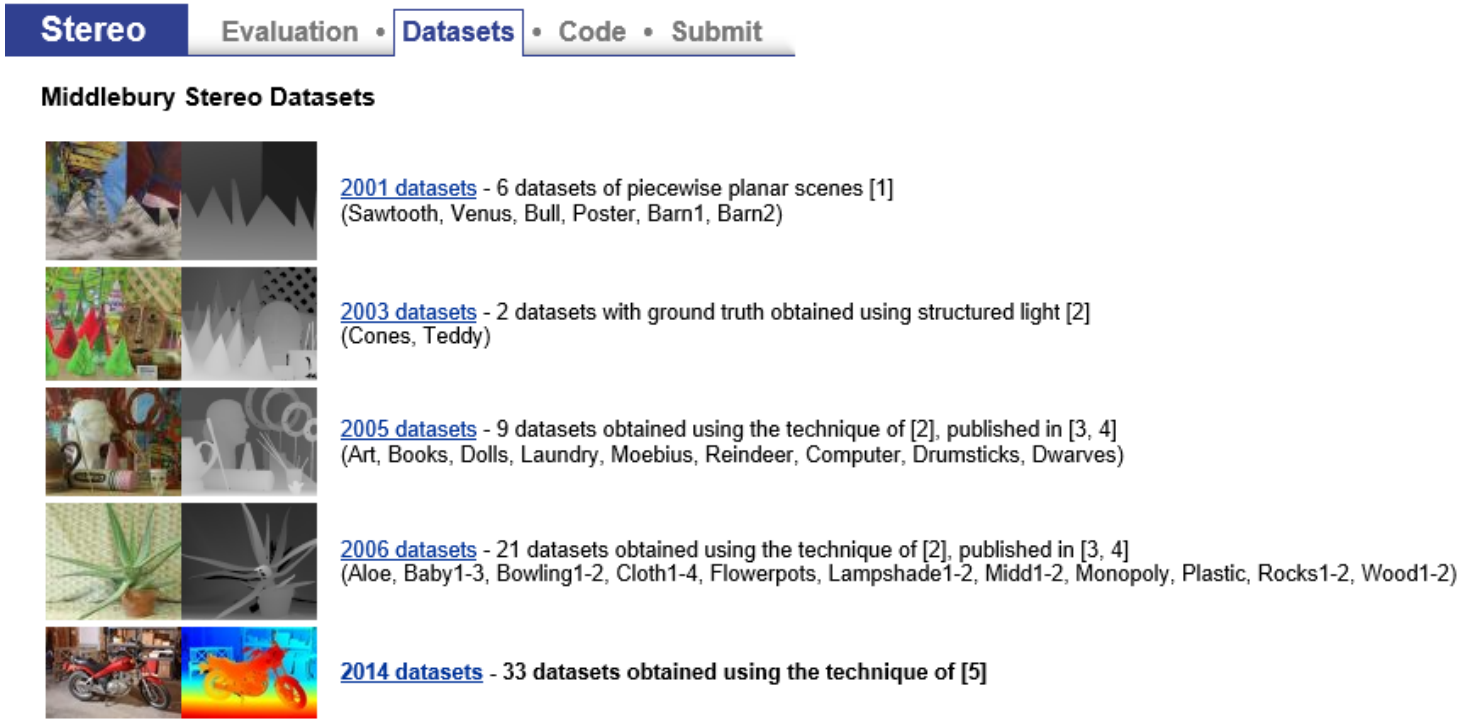
Similarity Measure	Formula
Sum of Absolute Differences (SAD)	$\sum_{(i,j)\in W} \lvert I_1(i,j) - I_2(x+i,y+j)\rvert$
Sum of Squared Differences (SSD)	$\sum_{(i,j)\in W} \left(I_1(i,j) - I_2(x+i,y+j)\right)^2$
Zero-mean SAD	$\sum_{(i,j)\in W} \lvert I_1(i,j) - \bar{I}_1(i,j) - I_2(x+i,y+j) + \bar{I}_2(x+i,y+j)\rvert$
Locally scaled SAD	$\sum_{(i,j)\in W} \lvert I_1(i,j) - \frac{\bar{I}_1(i,j)}{\bar{I}_2(x+i,y+j)} I_2(x+i,y+j)\rvert$
Normalized Cross Correlation (NCC)	$\frac{\sum_{(i,j)\in W} I_1(i,j).I_2(x+i,y+j)}{\sqrt[2]{\sum_{(i,j)\in W} I_1^2(i,j).\sum_{(i,j)\in W} I_2^2(x+i,y+j)}}$



SAD SSD NCC Ground truth

Stereo Vision

- Datasets
  - http://vision.middlebury.edu/stereo/data/



Thanks!